

Solar Adoption as Agricultural Adaptation: Evidence from California's Central Valley*

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Abstract

Does expansion of utility-scale solar energy threaten agricultural communities or offer them an economic lifeline? Using a parcel-level dataset spanning 2006-2023, we assess drivers of utility-scale solar adoption in California's Central Valley – one of the United States' most critical agricultural regions. We find that solar adoption concentrates in areas where agricultural viability is low and solar potential is high. Specifically, solar is most likely on parcels with poor soil, high farm labor costs, drought exposure, low groundwater levels, recent fallowing, large size, proximity to electricity infrastructure, and strong solar irradiance. A suitability index combining these factors strongly predicts adoption and identifies optimal sites that could generate 25.2 GW of solar energy, or 44% of California's 2045 target, on agriculturally marginal land parcels in the Central Valley. Results align with a conceptual framework in which landholders choose agriculture or solar leasing to maximize expected returns from their land while adapting to agricultural stressors.

Keywords: Solar Energy, Agriculture, Farmland Conversion, Water Scarcity, Renewable Energy Siting

JEL Codes: Q15, Q42, R14, Q54, Q24

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1 Introduction

The global transition to renewable energy is accelerating, with renewable energy capacity forecast to grow two to three times between 2024 and 2030 ([International Energy Agency 2024](#)). Among electricity generation options, utility-scale solar stands out due to its low levelized cost of energy ([Lazard 2025](#)). Despite this cost advantage and contributions to reducing carbon emissions and air pollution ([Rivera et al. 2024](#)), utility-scale solar raises concerns over the amount and type of land it replaces, with particular concern over displacement of farmland and trade-offs between clean energy and food production ([Winikoff et al. 2026](#); [Stid et al. 2025](#); [Maguire et al. 2024](#); [Battersby 2023](#); [Moore et al. 2022](#); [Hernandez et al. 2015](#)).

This study focuses on California’s Central Valley, an ideal location to examine trade-offs between agricultural land-use and solar energy development ([Stid et al. 2022](#)). The Central Valley supplies approximately 40% of the United States’ fruits, nuts, and vegetables and accounts for 8% of total U.S. agricultural output despite occupying just 1% of U.S. farmland ([USGS 2023](#)). However, characteristics that make the Central Valley an ideal location for agriculture, such as abundant sunlight, relatively flat land, and proximity to major population centers, also make it an ideal location for utility-scale solar projects. The number of these projects operating in the Central Valley increased from 0 in 2006 to 215 in 2023, with an average size of 52 hectares per project. Moreover, the Central Valley has experienced accelerating groundwater depletion since 2003 ([Liu et al. 2022](#)), jeopardizing agricultural viability and presenting utility-scale solar as a potential adaptation strategy for landholders.

Understanding which parcels convert to solar and why landholders make this choice is essential for managing tradeoffs between clean energy expansion and agriculture. These trade-offs are currently the focus of policy debates at national, state, and local levels. In 2025, the U.S. Department of Agriculture terminated grant and loan programs for solar development on farmland, citing concerns over impacts on agricultural communities and food security ([USDA 2025c](#)). At least 459 counties and municipalities imposed siting restrictions on renewables as of 2024 ([Eisenenson et al. 2025](#)), including over 800 ordinances on solar installations in effect, which reduce solar capacity in affected areas by 38% ([Lopez et al. 2023](#)). Despite the proliferation of such restrictions, comprehensive empirical evidence on the factors driving solar adoption remains limited, leaving policymakers without clear guidance on where solar

development is likely to occur, which agricultural lands are most vulnerable to conversion, and whether solar energy constitutes a threat or opportunity for rural communities. These questions are particularly policy relevant for California – where the Sustainable Groundwater Management Act is expected to force substantial fallowing of irrigated farmland by 2040 (Hanak et al. 2019; Escriva-Bou et al. 2023) – as well as other regions affected by drought, groundwater depletion, farm labor scarcity, and other threats to agricultural viability.

To analyze property-level determinants of utility-scale solar adoption, we construct a novel panel dataset encompassing the universe of non-urban land parcels in California’s Central Valley between 2006-2023.¹ For each parcel, we observe parcel size, solar radiation intensity, distance to electricity transmission infrastructure, soil suitability for agriculture (a measure that incorporates soil depth, surface texture, subsoil characteristics, and groundwater quality), irrigation well depth, and USDA Prime farmland classifications, as well as annual land use, agricultural labor costs, drought exposure, and the spatial footprint of utility-scale solar energy installations.

Using these granular data, we estimate the influence of parcel characteristics on landholders’ solar adoption decisions at both the extensive margin (solar = 0/1) and intensive margin (hectares installed). Because solar adoption is an absorbing state – once a parcel adopts, it remains adopted in every subsequent year – we restrict our estimation sample to parcel-years up to the year of adoption and estimate both a linear probability model and Cox proportional hazards model, reporting the two in parallel throughout. Our baseline specification includes both time-varying factors (i.e., lagged drought exposure and labor costs) and time-invariant parcel characteristics. We also estimate models with parcel fixed effects and interactions between time-invariant characteristics, drought exposure, and labor costs. We identify land use classes replaced by solar installations to quantify how much high-value farmland is being lost and assess which land uses are most likely to switch to solar. Finally, we extend our analysis to examine several additional dimensions of adoption: USDA farmland quality classifications (Prime, Unique, and Important farmland), neighbor effects (i.e., spatial spillovers) of solar adoption, irrigation stress as measured by groundwater well depth, and the role of land fallowing.

¹We observe 329,200 property boundaries for non-urban parcels. We exclude urban parcels because no utility-scale solar facilities were constructed in these areas during the study period.

Descriptive results reveal systematic patterns in solar siting. Solar facilities concentrate in areas with higher drought exposure, stronger solar irradiance, deeper groundwater well depth, and closer proximity to electrical substations compared to the overall parcel distribution. Solar siting is more dispersed across the agricultural suitability distribution: 69% of installations are on parcels in the bottom half of this distribution, while 24% are on parcels in the top two quintiles, representing high quality agricultural land. Solar facilities are under-represented in the highest-suitability deciles relative to the overall parcel distribution.

Between 2006 and 2023, cereal grains and fallow/idle cropland were the land types most often converted to solar, with 4,100ha (0.6% of total cereal grain area) and 3,833ha (0.8% of total fallow or idle cropland) converted, respectively. High-value permanent crops such as fruits and nuts experienced substantially lower conversion rates, consistent with landowners preserving sunk investments in perennial plantings. In total, 10,772 hectares in the Central Valley were converted to utility-scale solar during this period, representing 0.23% of the region's non-urban land area.

Regression results (consistent across LPM and Cox proportional hazards specifications) reveal that utility-scale solar adoption increases significantly with drought exposure, local agricultural labor costs, poor soil quality, solar irradiance, proximity to electrical substations, and parcel size. On the intensive margin, labor costs, solar irradiance, proximity to substations, and parcel size are all significant predictors of installed solar area among adopting parcels. Regressions including prior-period land use confirm that parcels with fallow/idle cropland, cereal grains, barren/shrubland, and developed open space are more likely to convert to solar relative to grassland/pasture, while high-value crops such as nuts, fruits, and feed, fiber, and industrial crops are significantly less likely to convert, consistent with landholders preserving investments in high-value perennial plantings.

We also explore several extensions. First, farmlands that are officially classified as Prime or Unique by the USDA are significantly less likely to convert to solar, while farmland classified as Important is more likely to convert, indicating that solar development avoids the highest-quality agricultural land but concentrates on land with intermediate agricultural value. Second, solar adoption likelihood on a parcel increases significantly following solar installation nearby, with the effect strongest within 2km and decaying with distance – consistent with peer learning or economies of scale or scope among neighboring landholders. Third, parcels

with deeper groundwater well depths – a proxy for irrigation stress – are significantly more likely to adopt solar. Fourth, solar adoption increases with the frequency of land fallowing in preceding years, indicating that solar often follows agricultural decline.

Finally, we construct a suitability index to identify non-solar parcels that are both technically attractive for solar development and economically marginal for agriculture. Using penalty functions that reward proximity to transmission infrastructure, large parcel size, high drought exposure, strong solar irradiance, poor soil quality, and low-value land uses, we identify 328 high-suitability sites (0.1% of total non-urban parcels) that could generate 25.2 GW of utility-scale solar energy, representing 44% of California’s 2045 renewable energy target (Weaver 2024). These optimal sites would concentrate solar development on marginal agricultural lands while meeting a substantial share of the state’s clean energy goals, demonstrating that renewable energy expansion and agricultural preservation are not in fundamental conflict. We validate the predictive power of our index by calibrating it on 2006-2018 data and predicting post-2018 adoption, finding that 87% of post-2018 parcels are installed in the top 3 deciles of predicted suitability.

To rationalize our results, we develop a conceptual framework in which a forward-looking parcel holder chooses whether to (i) continue with agriculture or (ii) allow a solar developer to lease their parcel to maximize expected returns from their land, accounting for their parcel’s unique characteristics and Bayesian updating based on prior drought exposure and labor costs. Together, our parcel-level empirical analysis and conceptual framework demonstrate that solar expansion in the Central Valley is not the result of random farmland displacement, but rather a systematic response by landholders to agricultural viability, solar economics, and infrastructure access.

1.1 Related Literature and Contributions

The rapid expansion of utility-scale solar energy generation has prompted a growing body of research on the determinants of solar adoption and the environmental trade-offs and impacts of these facilities. We contribute to four strands of this literature: (i) geospatial and economic drivers of solar siting, (ii) interactions between utility-scale solar and agriculture, (iii) policies and regulations governing solar siting, and (iv) landholder adaptation to climate shocks. We advance each of these literatures through a data-rich, parcel-level, longitudinal

analysis of solar adoption decisions in one of the world's most important agricultural regions.

Prior studies have identified physical and economic factors shaping the location of utility-scale solar development. Solar adoption is highest in areas with abundant solar radiation, low humidity, and moderate temperatures ([Adeh et al. 2019](#)), which often corresponds to croplands and grasslands. Furthermore, solar development has been found to disproportionately replace cultivated cropland, rather than brownfields or natural land cover – likely due to lower site preparation costs ([Evans et al. 2023](#); [Hernandez et al. 2015](#)). Proximity to electricity transmission infrastructure is critical for economic feasibility, as interconnection costs can exceed \$1 million per project if new transmission lines are required ([Lease 2022](#)). Institutional incentives also drive expansion. For instance, California's Renewable Portfolio Standard (RPS) and SB 100 mandate 100% renewable electricity by 2045 ([California Energy Commission 2021a,b](#)). As land competition intensifies, solar developers increasingly seek to optimize energy density and minimize land costs, leading them to prioritize larger parcels ([Bolinger and Bolinger 2022](#); [Daniels and Wagner 2022](#)). Our analysis leverages exceptionally detailed parcel-level geospatial data to quantify and compare the relative magnitude of these factors during a period of major solar expansion (2006-2023).

Another body of literature examines trade-offs between solar development and agriculture. Some farmers view solar leasing as a temporary land use that supports long-term agricultural viability, likening it to a 25-30 year “cover crop” ([Goldberg 2023](#)). “Agrivoltaics” offers a mixed land use strategy by co-locating solar and agriculture or livestock, though commercial viability remains limited ([Daniels 2023](#); [Pascaris et al. 2021](#)). Solar may also provide environmental co-benefits, such as water savings relative to irrigated cropland ([Stid et al. 2022](#)), making “solar fallowing” a potential drought adaptation strategy ([Buckley Biggs et al. 2022](#)). Empirically, [Winikoff et al. \(2026\)](#) find that solar is not disproportionately sited on Prime farmland nationally, but is more likely on farmland categorized as “Important” by state or local officials, with a slight negative correlation between overall farmland quality and solar development. [Maguire et al. \(2024\)](#) document that over 70% of rural solar farms were sited on agricultural land between 2009-2020, with approximately 15% of land surrounding solar installations shifting out of agriculture after solar development. Exploring tradeoffs between renewable development and irrigation, [Smith and Cooley \(2024\)](#) find that wind projects are less likely in areas with center pivot irrigation. [Abashidze and Taylor \(2022\)](#) find that so-

lar farms do not directly affect nearby agricultural land values. Policymakers have sought to steer solar development toward impaired or marginal lands ([Hoffacker et al. 2017](#); [CEQA 2024](#); [Kaenel and Venteicher 2024](#)), though most adoption decisions are still driven by individual landholder incentives. We advance this literature empirically and theoretically. Empirically, our novel parcel-level panel dataset allows us to incorporate time-varying drought exposure and farm labor costs – factors absent from prior cross-sectional studies – enabling analysis of how shocks to agricultural profitability drive solar adoption. Theoretically, we provide a conceptual framework formalizing landholders’ decision-making between agriculture and solar leasing under changing economic conditions.

Given extensive policy restrictions on utility-scale solar development across levels of government ([Eisensohn et al. 2025](#); [Lopez et al. 2023](#)), several studies explore regulatory dimensions of renewable energy siting, including [Fiechter et al. \(2024\)](#) and [Winikoff \(2022\)](#). [Li et al. \(2025\)](#) find that a policy promoting renewable energy increased land values for solar-suitable parcels near electrical substations, while [Jarvis \(2025\)](#) find that renewable installations have heterogeneous impacts on nearby property values and planning officials are highly sensitive to local costs. We contribute to this literature by showing that landholders disproportionately adopt solar on declining or marginal agricultural lands, suggesting that policymakers could guide solar siting toward these lands rather than restricting development outright.

Finally, there is a growing literature on how landowners adapt to climate shocks, including drought ([Ortiz-Bobea et al. 2021](#); [Burke and Emerick 2016](#)). [Chandanpurkar et al. \(2025\)](#) document continental drying and mega-drought regions across the Northern Hemisphere, and [Liu et al. \(2022\)](#) document accelerating groundwater depletion in California’s Central Valley. In response, farmers in the Central Valley have shifted toward higher-value perennial crops while reducing lower-value annual crops ([Mount and Hanak 2019](#); [Escriva-Bou et al. 2023](#); [Hanak et al. 2019](#); [Carman 2019](#)). Water scarcity has also driven crop switching and land fallowing ([Medellín-Azuara et al. 2012](#); [Wartenberg et al. 2021](#)). Wealthier landowners are better positioned to adapt to drought, creating unequal distributional effects ([Visser et al. 2024](#)). We extend this literature by documenting solar adoption as a potential climate adaptation strategy. We show that drought exposure is significantly positively associated with solar adoption, particularly on parcels with low agricultural viability. Further, we show that solar adoption likelihood increases with fallow duration and groundwater well depth, in-

dicating that solar leasing offers a revenue-generating option for landholders struggling to sustain agricultural profitability. These findings position solar adoption as a stabilizing force for rural communities under mounting climate pressures.

2 Conceptual Framework

Suppose that parcel owners face a forward-looking land use decision: whether to continue using their land for agriculture or lease it to a solar energy developer. If solar panels are installed, the land is removed from agricultural production and generates a stream of lease payments for approximately 25 years. From the parcel owner’s perspective, this choice involves comparing the net present value (NPV) of two mutually exclusive options:²

- **Agricultural use**, which yields uncertain long-term profits depending on evolving market prices and drought conditions
- **Solar leasing**, which promises fixed, inflation-adjusted payments over a 20 to 30-year horizon with no labor or input costs.³

Let $D_i \in \{A, S\}$ denote the land use choice for parcel i , where A represents agriculture and S solar leasing, and let T denote the duration of a typical solar lease.⁴

²Agrivoltaics, or the co-location of agriculture and solar panels, allows landholders to combine agricultural production with solar energy generation rather than treating them as mutually exclusive uses. In practice, this can involve grazing sheep to manage vegetation under and around panel arrays, or cultivating low-growing, shade-tolerant crops such as lettuce, strawberries, or peppers beneath elevated panels. Despite their potential to reduce land use tradeoffs, agrivoltaic systems have yet to gain traction in California’s Central Valley, with only 2 of the 215 cited solar facilities currently reporting agrivoltaic implementation as of 2023.

³In practice, solar lease payments are not fully guaranteed. Landholders signing long-term contracts with solar developers face risk from developer bankruptcy or sale of their contract to another company that seeks to renegotiate it on less favorable terms, damage, degradation, or technological obsolescence of their panels, disputes over decommissioning obligations at the end of the lease, or potential tax liability if the developer fails to cover the switch from agricultural to commercial property status. For simplicity, we do not formally model these risks. In principle, they would enter solar returns as a discount factor on the effective value of each lease payment, with landholders updating their beliefs about this risk factor based on observed outcomes at nearby solar installations. Furthermore, a parcel’s suitability for solar is not captured by size, solar irradiance, and proximity to grid infrastructure alone. Local zoning restrictions, transmission congestion, interconnection queues, and other factors can all play a role. We abstract away from these unobserved factors in our framework and analysis for simplification.

⁴A third option available to some landholders is sale for urban development. Like solar leasing, land sale for development does not require ongoing labor or input costs and avoids exposure to future agricultural risk. Unlike solar leasing, its payoff arrives as a single upfront sum rather than a stream of payments, and its value depends on a different dimension of land suitability – proximity to and desirability for urban expansion – rather than solar radiation or grid access. We omit urbanization from our conceptual framework to focus on the tradeoff between agriculture and solar, which is the central interest of this paper, and because our empirical sample excludes parcels that are ever urban during the study period. Utility-scale solar is never installed on urban parcels during our study period, suggesting these areas lie outside solar developers’ choice set.

2.1 Agricultural Returns

Let $\mathbb{E}[\pi_{it}^A]$ denote the landholder's expected annual net return from agriculture on parcel i in future period $t + \tau$. Expected agricultural returns depend on parcel characteristics, current land use, and beliefs about future drought exposure and agricultural labor costs:

$$\mathbb{E}_t[\pi_{i,t+\tau}^A] = f(\text{Suitability}_i, \text{CurrentUse}_{it}, \mathbb{E}_t[d_{i,t+\tau}], \mathbb{E}_t[w_{i,t+\tau}]) \quad (1)$$

where d_{it} denotes drought exposure and w_{it} denotes farm labor costs. Function $f(\cdot)$ is hypothesized to be increasing in *agricultural suitability*, dependent on *current land use* (because parcels with pasture or temporary crops may be more likely to adopt solar than those with permanent crops), and decreasing in *expected drought exposure* and *expected farm labor costs*.

Bayesian Learning We assume landholders form expectations about d_{it} and w_{it} through Bayesian updating based on recent observed conditions. Specifically, suppose realized drought exposure and labor costs can be decomposed into an unknown parcel-specific mean component and transitory shocks:

$$d_{it} = \mu_{di} + \varepsilon_{it}^d \quad w_{it} = \mu_{wi} + \varepsilon_{it}^w \quad (2)$$

where μ_{di} and μ_{wi} are unknown long-run mean conditions for parcel i , and ε_{it}^d and ε_{it}^w are mean-zero shocks. At time t , the landholder observes $(d_{i,t-1}, w_{i,t-1})$ and updates beliefs about (μ_{di}, μ_{wi}) . Under conjugate priors, the posterior mean is a convex combination of the prior mean m and the most recent realization:

$$\mathbb{E}_t[\mu_{di}] = (1 - \omega_d)m_{d0} + \omega_d d_{i,t-1} \quad \mathbb{E}_t[\mu_{wi}] = (1 - \omega_w)m_{w0} + \omega_w w_{i,t-1} \quad (3)$$

for weights $\omega_d, \omega_w \in (0, 1)$ that summarize the relative precision of prior beliefs and the informativeness of observed conditions. We assume expectations of future drought exposure and labor costs are anchored by these posterior means, implying that higher recent drought exposure or labor costs increase expected future exposure and costs, lowering expected agricultural returns through function $f(\cdot)$ in Equation (1).

Agricultural net present value. Under the simplifying assumption that agricultural land retains productive value over time,⁵ we define the time- t present value of continuing in agriculture over the duration T of a typical solar lease, given discount factor $\delta \in (0, 1)$, as:

$$\mathbb{E}_t[V_{it}^A] = \sum_{\tau=0}^{T-1} \delta^\tau \mathbb{E}_t[\pi_{i,t+\tau}^A]. \quad (4)$$

In our empirical analysis, lagged drought exposure and labor costs on parcel i in the previous three years proxy for the information used to form $\mathbb{E}_t[d_{i,t+\tau}]$ and $\mathbb{E}_t[w_{i,t+\tau}]$.

2.2 Solar Leasing Returns

Let R_{i0}^S be the initial lease payment offered by a solar developer for parcel i . Contracts typically include an annual escalation factor γ (e.g., 2% per year) to adjust for inflation. The total lease payment in year t is:

$$R_{it}^S = R_{i0}^S \cdot (1 + \gamma)^t$$

The NPV of a T -year solar lease is:

$$V_i^S = R_{i0}^S \cdot \sum_{t=0}^{T-1} [\delta(1 + \gamma)]^t = R_{i0}^S \cdot \frac{1 - \theta^T}{1 - \theta} \quad \text{where } \theta = \delta(1 + \gamma) < 1$$

The lease rate R_{i0}^S depends on:

$$R_{i0}^S = g(\text{SolarRadiation}_i, \text{GridDistance}_i, \text{Size}_i)$$

This function is hypothesized to be increasing in *solar irradiation*, decreasing in *distance from electricity grid infrastructure*, and increasing in *parcel size*, given that more solar radiation generates more electricity, long-distance grid connections are costly, and larger parcels enable economies of scale in permitting, installation, and operations – increasing solar developers' willingness to pay for these characteristics.

⁵If sustained cultivation depletes soil fertility or groundwater stocks while “solar fallowing” allows regeneration during the lease term, this would provide an additional incentive for solar adoption not captured in the baseline framework.

2.3 Parcel Owner’s Decision Rule

Notably, expected returns from agriculture depend on uncertain future outcomes, while returns from solar depend on observable baseline characteristics. A risk-neutral landowner⁶ thus chooses the option with higher NPV at time t :

$$D_{it} = \begin{cases} S & \text{if } V_{it}^S > \mathbb{E}_t[V_{it}^A] \\ A & \text{otherwise,} \end{cases}$$

This framework predicts that landholders are more likely to adopt solar when expected agricultural profits are low (e.g., due to poor soil, low groundwater levels, high labor costs, or drought exposure), solar potential is high (e.g., due to strong solar radiation or proximity to grid infrastructure), and parcel size is large (enabling cost efficiencies in solar installation and operations). Empirical calibration of each parameter in agricultural and solar return functions $f(\cdot)$ and $g(\cdot)$ is undertaken in the following sections.

3 Data

This section discusses data sources and processing steps to prepare the parcel-level panel dataset (2006-2023) used in empirical analyses.

3.1 Parcels

Parcels serve as the primary unit of analysis. The base parcel boundary layer (as of 2013) is drawn from the California Statewide Parcel Boundaries dataset ([LACounty 2025](#)). Missing counties are supplemented using county-level GIS portals. Parcel data do not report information on owners, limiting our ability to analyze the effects of owner characteristics on solar adoption. To focus the analysis on land where solar development is feasible, we exclude parcels identified as urban in the National Land Cover Dataset (NLCD) as of 2023, given that

⁶This risk-neutral framework can be extended to incorporate risk aversion. For a landowner with constant absolute risk aversion parameter $\rho > 0$, the certainty equivalent of perpetual agricultural returns is $CE(V_{it}^A) = \frac{1}{1-\delta} [\mathbb{E}[\pi_{it}^A] - \frac{\rho}{2}\sigma_{itA}^2]$, where σ_{itA}^2 is the variance of annual agricultural profits. Risk aversion parameter ρ could vary across landholders based on individual preferences, wealth, access to crop insurance, off-farm employment opportunities, or other risk-hedging mechanisms. Under this framework, parcels with high agricultural profit volatility would face a higher risk premium $\frac{\rho}{2}\sigma_{itA}^2$, providing additional motive for solar adoption.

no utility-scale solar facilities were built in urban areas during the study period. The resulting parcel boundaries, shown in Figure 1, are intersected with geospatial datasets described in the following sections to construct the parcel-year panel.

Figure 1: Non-Urban Parcels

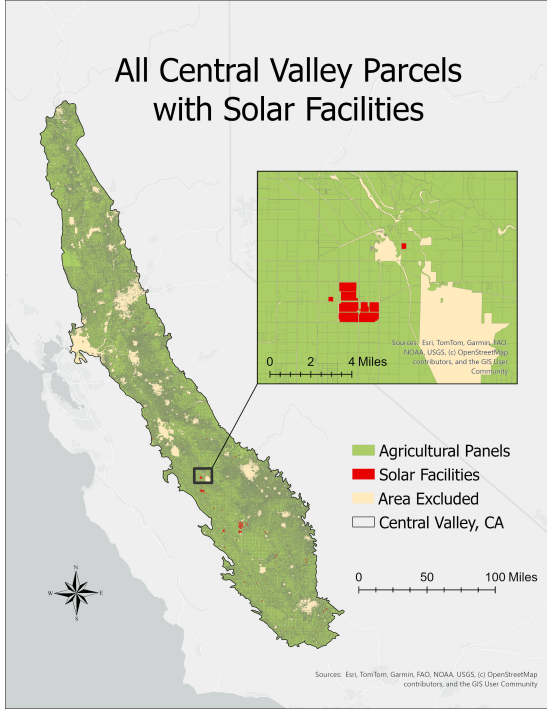
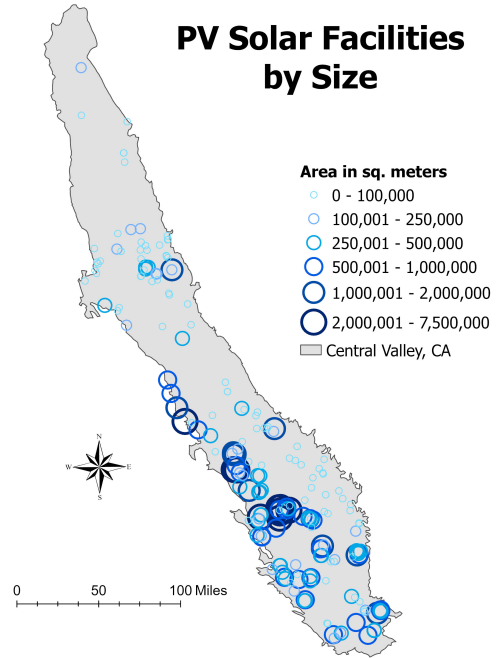


Figure 2: Solar Facilities by Size



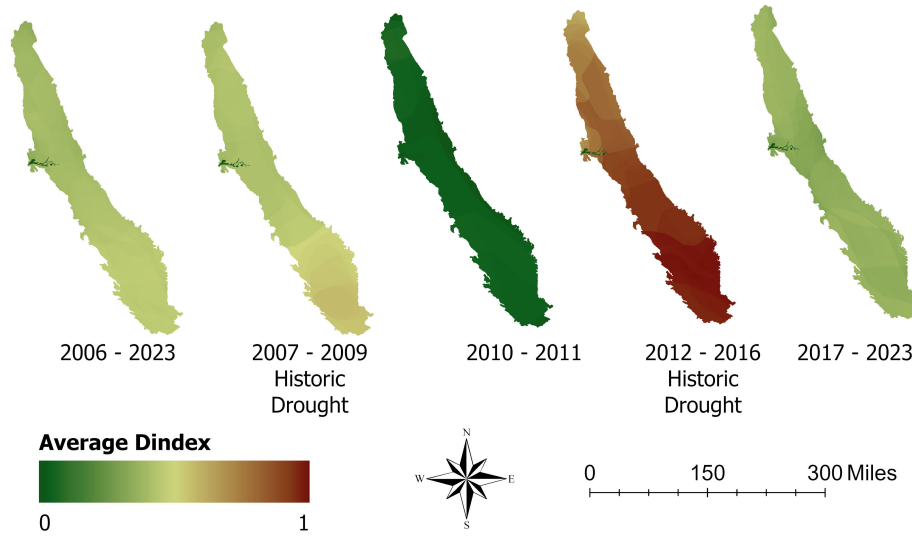
3.2 Solar Facilities

Data on utility-scale solar installations are from the U.S. Solar Photovoltaic Database (Fujita et al. 2023). This dataset includes year of installation, facility size (including ancillary infrastructure such as substations or maintenance facilities), and generating capacity for all utility-scale solar facilities with more than 1MW of capacity installed in the United States between 2006-2023. 215 solar facilities are recorded within the study region during this period. Parcels are coded as having a solar facility in year t if any portion of a utility-scale solar site falls within the parcel boundary in that year. Some installations cross parcel boundaries and are thus recorded for multiple parcels. Total solar-covered area is measured in square meters per parcel-year. Figure 2 presents the location and relative size of these facilities.

3.3 Drought

Drought exposure is measured using the U.S. Drought Monitor (USDM) ([University of Nebraska–Lincoln 2025](#)), which provides weekly maps of drought severity across the United States at 500m resolution. The USDM integrates multiple physical indicators, including precipitation, streamflow, reservoir levels, temperature, soil moisture, vegetation health, and expert field assessments. Drought is categorized into five levels: Abnormally Dry (D0), Moderate (D1), Severe (D2), Extreme (D3), and Exceptional (D4). Based on these scores, the Drought Severity and Coverage Index (DSCI) ([Akyuz 2017](#)) constructs a continuous measure of drought severity. DSCI assigns integer weights to each category each week: 1 to D0, 2 to D1, through 5 to D4, yielding a maximum possible annual score of 500. We normalize DSCI scores to a 0–1 scale. Figure 3 presents average annual drought intensity over time, with years of declared historic drought highlighted.

Figure 3: Drought Exposure Over Time

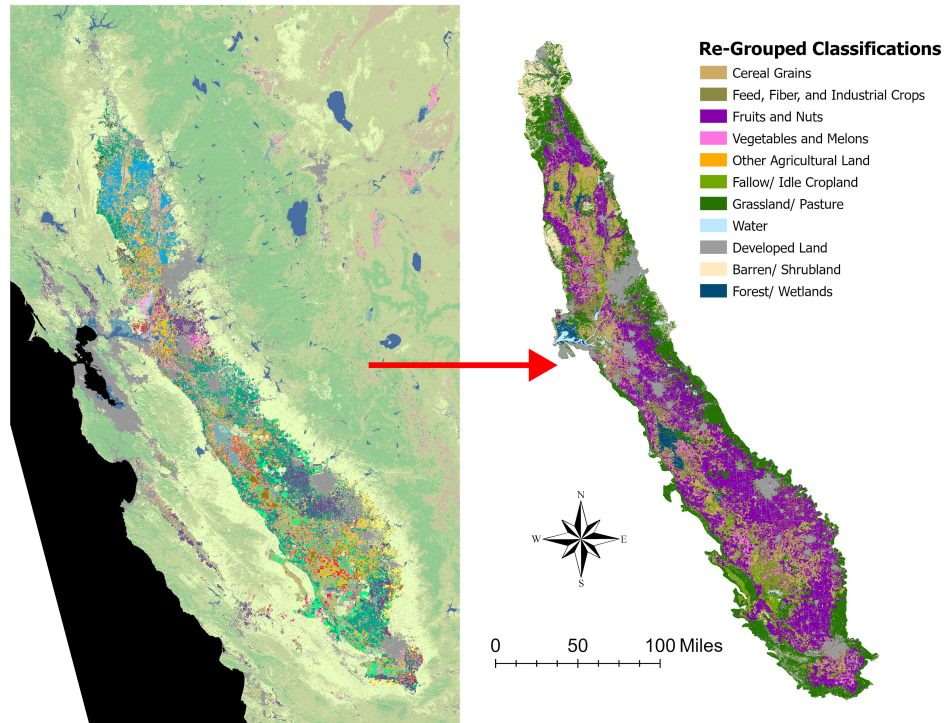


3.4 Crops and Land Use

Annual crop coverage for 2007 through 2023 is drawn from the [USDA National Agricultural Statistics Service \(2023\)](#), which provides land cover information at a 30m-by-30m resolution. We re-classify land uses into 12 groups: Cereal Grains; Feed, Fiber, and Industrial Crops; Fruits; Vegetables and Melons; Nuts; Tree Crops; Fallow /Idle Cropland; Grassland/Pasture;

Water; Developed Land; Barren/Shrubland; and Forest/Wetlands. The breakdown of these categories is listed in Table A1. Figure 4 illustrates the reclassified layer for 2022 as a selected example. Because 2007 is the first year where reliable crop information is recorded for California’s Central Valley, installation of a single solar facility in 2006 is omitted from the analysis. We identify the land use for each parcel-year as the modal land use in that year, enabling identification of specific crop types displaced by solar development.

Figure 4: Reclassified Crop Data Layer (2022)



3.5 Agronomic Suitability

Agronomic suitability data are obtained from the [Conservation Biology Institute \(2024\)](#), which processes soil information in the Soil Survey Geographic Database (SSURGO) ([USDA 2022](#)) to generate an agricultural potential index. The index incorporates factors such as soil depth, surface texture, subsoil composition, and groundwater quality to estimate agronomic potential, expressed as a percentage. We rescale this percentage to a 0 to 1 range and invert it so that higher values indicate poorer suitability for agriculture. The SSURGO dataset does not cover the full extent of the Central Valley, resulting in the exclusion of 27 solar facilities.

Excluded areas are shown in gray in Figure 5. We estimate a robustness check where we drop the agricultural suitability variable to retain all solar facilities in sample, shown in Table A2, and find that estimates for the remaining covariates are stable.

Figure 5: Agricultural Suitability

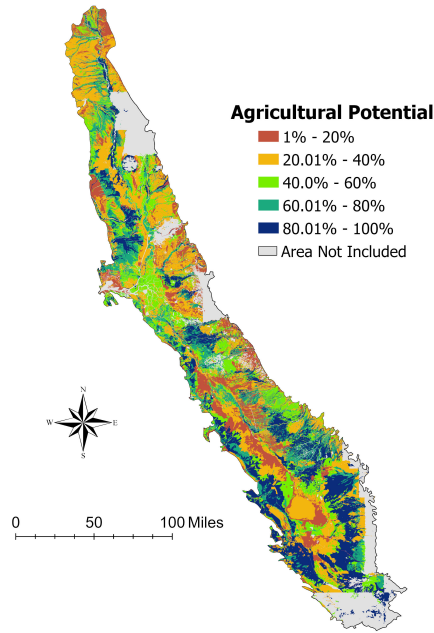
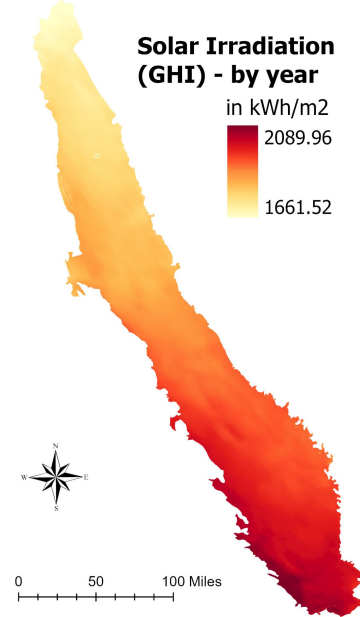


Figure 6: Solar Irradiation



3.6 Solar Irradiance

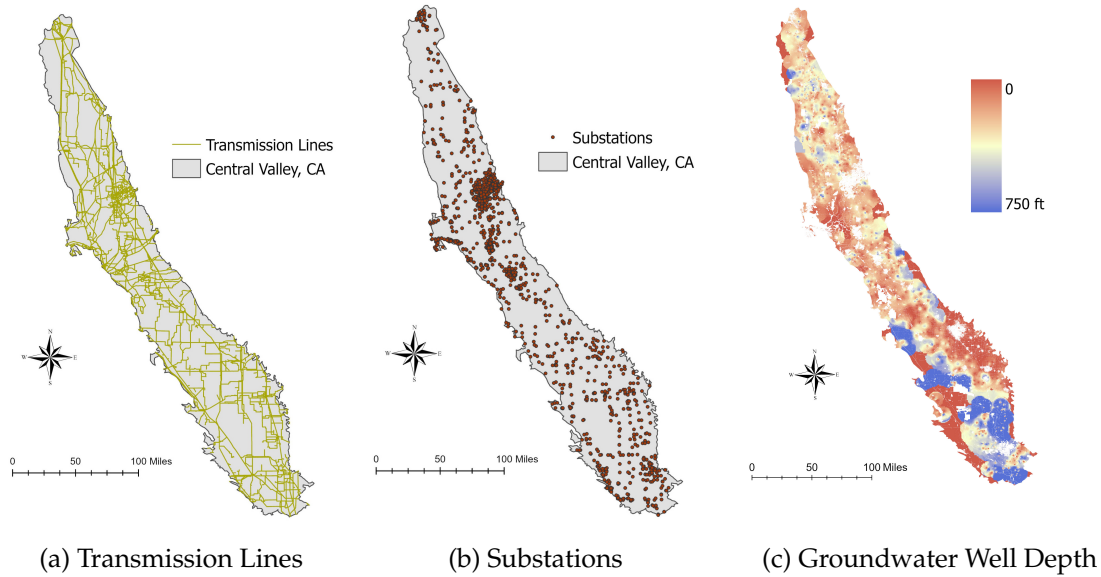
Average annual solar potential is collected from the Global Solar Atlas ([Solargis 2023](#)), which reports Global Horizontal Irradiance (GHI) in kWh/m² at a 250m resolution, as shown in Figure 6. We normalize GHI to a 0–1 scale to facilitate comparison across parcels. GHI measures the total solar radiation received on a horizontal surface, and is a standard metric for evaluating the energy potential of solar installations. Notably, the average GHI value in the Central Valley is 1,952 kWh/m², exceeding the national average of 1,599 kWh/m².

3.7 Energy Infrastructure

Locations of electricity transmission lines and substations are from the [Conservation Biology Institute \(2024\)](#). This dataset maps major energy infrastructure in California as of 2022

(Figure 7a-b). Distances to transmission lines and substations, plotted in Appendix Figure A1, are calculated as the Euclidean distance from each parcel centroid to the nearest infrastructure point. We invert and log these distances so higher values indicate closer proximity.

Figure 7: Electricity Transmission Infrastructure and Groundwater Depth



3.8 Farm Labor Costs

We compile data on farm workers' average weekly wages at the county-year level from the Quarterly Census of Employment and Wages ([Bureau of Labor Statistics 2024](#)). We define farm workers using NAICS code 115115 (Farm Labor Contractors and Crew Leaders) and ascribe prevailing wages to all parcels within that county-year, approximating the costs parcel-holders would face when hiring agricultural workers in their local labor market. We deflate nominal wages using the GDP deflator from FRED, and plot average wages between 2006-2023 as shown in Appendix Figure A2.

3.9 Supplementary Data

Prime, Unique, and Important farmland classifications are drawn from the California Farmland Mapping and Monitoring Program ([CA Dept of Conservation 2026](#)). Prime farmland possesses ideal physical and chemical properties to sustain high-yield agricultural pro-

duction, and must have been recently irrigated for agricultural production. Unique farmland has lower-quality soils but has recently been planted with high-value crops such as fruits or nuts, and is typically irrigated. Important farmland combined Statewide Important Farmland, which has minor shortcomings such as sloped terrain or lower soil quality, but has recently been irrigated, and and county-designated Locally Important Farmland (USDA 2026). Rural-Urban Continuum Codes measuring county-level urbanization in 2003 and 2023 are from USDA (2025b). Electrical substation locations as of 2014 (used in a robustness exercise) were obtained through a data request to the California Energy Commission (CA Energy Commission 2026). Groundwater well depth is constructed from the Periodic Groundwater Level Measurements dataset (CA Dept of Water Resources 2026a,b), which combines depth-to-groundwater readings from several agencies and programs, retaining only verified readings from wells with at least 10 years of consistent monitoring. We assign each parcel a well depth using inverse distance weighting across nearby monitoring wells. Parcel-level groundwater well depths are mapped in Figure 7c. Because reliable well depth measurements are unavailable for roughly one-third of parcels in our sample, we omit this variable from our main analysis and instead examine it in supplementary analyses using the subset of parcels with well-depth data.

4 Empirical Strategy

We estimate panel regression and survival models to quantify the effects of parcel characteristics on utility-scale solar adoption. In our baseline panel specification, we regress solar adoption in parcel p and year t on three-year lagged averages of drought severity ($D_{p,t-1:t-3}$) and agricultural labor costs ($L_{p,t-1:t-3}$) and a vector $\mathbf{X}_p$ of time-invariant parcel characteristics (agronomic suitability, solar radiation intensity, proximity to electrical substation and transmission lines, and parcel area), including year fixed effects (δ_t) and clustering standard errors at the parcel level:

$$\text{SolarAdoption}_{pt} = \beta D_{p,t-1:t-3} + \gamma L_{p,t-1:t-3} + \mathbf{X}_p' \mu + \delta_t + \epsilon_{pt} \quad (5)$$

$\text{SolarAdoption}_{pt}$ denotes either (i) a binary indicator for whether a new solar facility was installed in parcel p in year t (assessing the extensive margin of adoption), or (ii) the total

solar-covered area in m² (assessing the intensive margin).⁷ Given that solar installation is effectively irreversible within our sample period, once *SolarAdoption_{pt}* switches to 1 for a parcel, it remains 1 in every subsequent year. These repeated 1s do not reflect new adoption decisions and mechanically inflate the share of adopter-years in the estimation sample. We correct for this over-weighting by dropping parcel-years after adoption. We estimate this corrected panel specification using OLS, and re-estimate it (for binary outcomes) using logit as an additional robustness check. Agricultural labor costs (real USD), inverse distances (km) and parcel area (hectares) are logged.

As a complementary approach, we use a Cox proportional hazards model for the time until parcel p transitions to solar use:

$$h_p(t \mid D_{p,t-1:t-3}, L_{p,t-1:t-3}, \mathbf{X}_p) = h_0(t) \exp \left(\beta D_{p,t-1:t-3} + \gamma L_{p,t-1:t-3} + \mathbf{X}_p' \mu \right) \quad (6)$$

where $h_p(t \mid \cdot)$ is the hazard of adoption for parcel p in year t , conditional on not having adopted in any prior year, and $h_0(t)$ is a baseline hazard common to all parcels. We cluster standard errors at the parcel level and report hazard ratios ($\exp(\hat{\beta})$). Parcels that have not adopted solar by the end of the sample period (2023) are right-censored. As an additional robustness check, we estimate a complementary log-log (cloglog) model, the discrete-time analog to the continuous-time proportional hazards model.

Given that explanatory variables are measured in different units, direct comparison of coefficient magnitudes is not meaningful. We therefore standardize each continuous covariate by subtracting its mean and dividing by its standard deviation prior to estimation, and report standardized coefficients for both the corrected LPM and Cox specifications, allowing comparison of relative importance both within and across specifications.

We also examine several extensions of the baseline models. First, we test whether drought exposure and labor costs interact with other parcel characteristics to predict solar adoption, with and without parcel fixed effects (in the latter case omitting time-invariant covariates since these are absorbed by the fixed effects). Second, we examine how prior land cover shapes solar adoption by including a set of land-use dummies capturing the modal land

⁷We include lagged drought exposure and farm labor costs to test the hypothesis that parcel-holders may update their expectations of future agricultural viability based on their experience of drought and labor costs in prior years. Agriculture in the Central Valley is typically labor intensive, while solar does not require any labor inputs from the parcel-holder. We include three years of lag to account for the typical timeline required to plan, permit, and construct solar facilities (Richardson 2023). We test 1- and 5-year lags in a robustness check.

cover on parcel p in year $t - 1$, i.e., the type of land cover replaced by solar on adopting parcels. Grassland and pasture – a standard option for solar siting – serves as the omitted reference category. Third, we add indicators for USDA-classified Prime, Unique, and Important farmland to test whether solar development disproportionately targets the highest-quality agricultural land. Fourth, we assess spatial spillovers by including lagged measures of neighboring solar development within 2, 5, and 10km radii. Fifth, we estimate the effects of irrigation access by re-estimating models supplemented with groundwater well depth as an additional covariate. Finally, we construct measures of fallowing intensity to test whether solar adoption is more likely on lands with declining agricultural viability.

4.1 Identification

Our empirical analysis serves primarily as a descriptive characterization of how solar adoption correlates with parcel characteristics and local conditions, revealing systematic patterns in solar siting. Several features of our data and methods reduce concerns that these correlations reflect reverse causality, simultaneity, or other confounding. First, year fixed effects (δ_t) absorb time-varying factors common across parcels, including state and federal policy changes and evolving solar technologies. Parcel fixed effects (η_p) absorb time-invariant parcel-level factors, including unobserved owner characteristics insofar as these remain stable over time. Second, time-varying covariates are plausibly exogenous at the parcel-level: drought exposure is determined by regional climate patterns beyond any landholder’s control, and labor costs are set in regional labor markets and measured at the county level, making individual parcel-holders price takers.⁸ Prior land use, in contrast, is time-varying but

⁸One concern is that drought exposure and farm labor costs may vary primarily across years rather than across parcels, in which case our drought and labor cost coefficients could be coincident with a common year effect. We plot average trends in drought exposure and farm labor costs in Appendix Figure A3, revealing that drought varies substantially across years, with intense spikes in 2007-2009, 2012-2016, and 2020-2022, while real labor costs have increased steadily from around \$350/week in 2006 to around \$550/week in the 2020s. We assess within-year spatial variation in drought exposure and farm labor costs in Appendix Figures A4 and A5. For drought exposure, year fixed effects explain 94.0% of total variance; of the remaining 6.0%, 14.9% reflects a persistent, parcel-specific component, while 85.1% is idiosyncratic variation across parcels and over time. For farm labor costs, year fixed effects explain 57.0% of total variance. Of the remaining 43.0%, 62.2% reflects a persistent county-specific component, while the rest is idiosyncratic year-to-year variation. Although each variable’s residual, within-year variation is modest as a share of total variance, our sample spans millions of parcel-year observations for drought and hundreds of county-years for labor costs, providing sufficient cross-parcel variation within each year to identify effects. Nevertheless, we recognize that drought effects are identified off limited within-year variation, are sensitive to model choice, and lose statistical significance once we include parcel or county fixed effects. Drought effect estimates should be interpreted with these caveats in mind.

chosen at the parcel level, making this covariate more liable to endogeneity. Third, time-invariant covariates are not affected by individual landholder behaviors. Agronomic suitability is measured at a spatial resolution substantially larger than individual parcels, making it unresponsive to parcel-level management practices. Distance to transmission infrastructure reflects placement of major substations and transmission lines based on grid-level considerations, and results are robust to using substation locations from 2014 rather than 2022 to avoid potential reverse causality. Groundwater well depth is registered at test wells and applied (using inverse distance weighting) for all nearby parcels, such that individual landholders' irrigation decisions do not directly affect well depth measurements. Parcel size is fixed over our study period. While landholders may have selected into time-invariant parcel characteristics endogenously, the rapid evolution of solar technology between 2006 and 2023 means utility-scale solar was not available for most parcel-holders during the early study period. Landholders' initial parcel and crop choices were thus likely taken when solar was not in the feasible choice set, reducing concerns over anticipatory selection into favorable solar sites. Omitted variable bias is minimized by our extensive set of covariates.

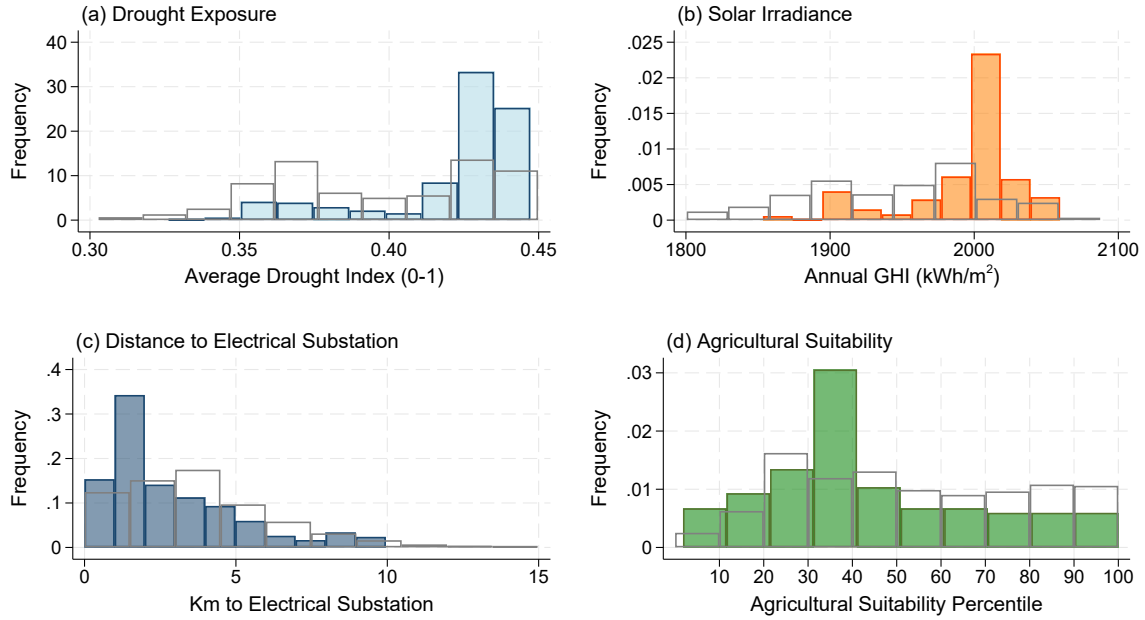
5 Results

5.1 Descriptive Results

There were 215 utility-scale solar energy projects installed across 419 parcels in California's Central Valley between 2006-2023, averaging 52 hectares per project. Figure 8 plots histograms of solar installations across deciles of parcels' (a) drought exposure, (b) solar irradiance, (c) distance to the nearest electrical substation, and (d) agricultural suitability (i.e., soil quality). Histograms for non-solar parcels are plotted in gray outlines for comparison.

As shown in Figure 8a, solar installations are strongly clustered on parcels with higher average drought exposure between 2006-2023. 71% of solar sites are in the upper quintile of drought severity, with a further 12.2% in the fourth quintile. Non-solar parcels are much more evenly spread across drought exposure. Figure 8b shows that 79% of solar installations are located on parcels in the top two quintiles of solar irradiance. The left tail of solar sites on below-median parcels reflects the fact that solar irradiance throughout the Central Valley exceeds the national average, making solar panels economically viable even in relatively

Figure 8: Distribution of Solar Siting Across Parcel Characteristics



Notes: Panels a-d plot histograms of 419 parcels hosting utility-scale solar facilities (colored bars) compared to 328,781 non-solar parcels (gray outlines) across four parcel characteristics. Panel (a): Average drought exposure index across the 2006-2023 period, normalized to a 0-1 scale. Panel (b): Annual global horizontal irradiance (GHI) in kWh/m². Panel (c): Distance to nearest electric substation in kilometers. Panel (d): Agricultural suitability measured as percentiles, with higher percentiles indicating higher productivity.

low-irradiation areas. Figure 8c reveals that distance to an electrical substation is a strong determinant of solar siting: 49% of installations are within 2km from a substation, 90% are within 5km, and all are within 10km (mean distance = 2.9km). In contrast, the mean non-solar parcel is 4.1km from a substation. Finally, Figure 8d documents that solar installations are located across the full distribution of agricultural suitability: 69% are on parcels in the bottom half of the suitability distribution, while 31% are on parcels in the top half – including 24% on parcels in the top two quintiles, representing high quality agricultural land. However, compared to the distribution of non-solar parcels, solar installations are relatively less concentrated on the most productive land.⁹

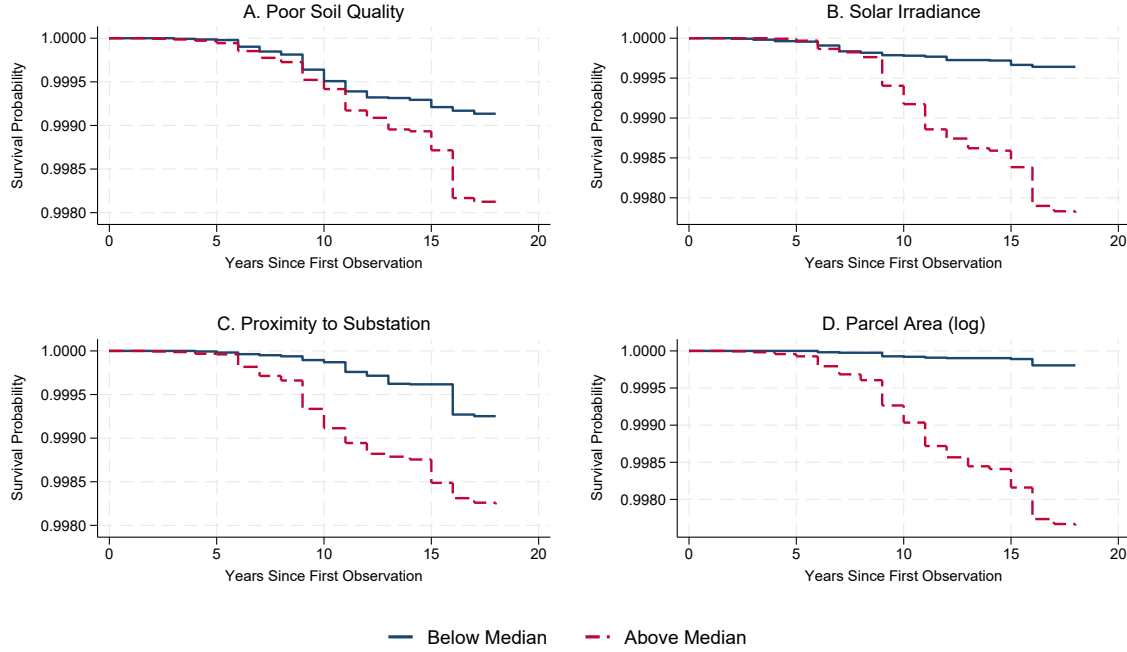
Because solar adoption is best understood as a time-to-event process rather than a series of independent yearly draws, we next characterize adoption patterns descriptively using

⁹Appendix Figure A6 reports an analogous histogram of solar installations by groundwater well depth, showing that installations disproportionately cluster on parcels with deeper wells.

Kaplan-Meier survival curves, which plot the probability that a parcel has not yet adopted solar, split by whether each characteristic is above or below its sample median (Figure 9). Parcels with poor soil quality (9a), high solar irradiance (9b), proximity to an electrical substation (9c), and large size (9d) show markedly lower survival, i.e., faster and more frequent solar adoption, relative to their below-median counterparts. Appendix Figure A7 reports analogous curves for Prime, Unique, and Important farmland designations. Consistent with regression results presented below, Prime farmland parcels show substantially slower adoption than non-Prime parcels, while Important farmland parcels show faster adoption than non-Important parcels. Unique farmland parcels show slightly higher survival than non-unique parcels, but the separation is small. Finally, Appendix Figure A8 reports Kaplan-Meier curves for parcels in the 0-50th percentile, 50-75th percentile, 75-90th percentile, and 90-100th percentile of groundwater well depth. Parcels with relatively shallow well depths (0-75th percentile) show very low solar adoption rates, parcels with deeper wells (75-90th percentile) exhibit moderate adoption rates, and parcels with the deepest wells (90-100th percentile) exhibit much higher adoption rates, suggesting that difficulty in accessing groundwater for irrigation encourages solar adoption.

Figure 10 reports land area converted to solar between 2007 and 2022 by land use category. The largest conversions occurred from cereal grains (4,100 ha, or 0.6% of total cereal grain area), fallow and idle cropland (3,833ha, or 0.8% of total fallow or idle cropland area), grassland and pasture (1,083 ha), and feed, fiber, and industrial crops (715 ha). Fruits, nuts, vegetables and melons, and especially tree crops are substantially less likely to be converted to solar – consistent with landowners prioritizing the conversion of flexible or low-return acreage for solar development while retaining more profitable crops and sunk investments. Land conversion to solar was negligible prior to 2010, after which solar expanded rapidly, with a peak in 2015 at 1,845 total hectares of land converted. Solar installations slowed over subsequent years before rebounding in 2019 (2,002 ha) and 2020 (2,248 ha). In total, 10,772 ha of land in the Central Valley were converted to solar during the study period: 5,558 ha from actively cultivated agricultural land, 4,947 ha from grassland, pasture, or fallow/idle cropland, and 267 ha from other land uses (e.g., developed land, forest, wetlands). By 2022, solar facilities occupied 0.23% of non-urban land in the Central Valley. Numeric values of land converted to solar per crop type-year are detailed in Appendix Table A3. Appendix Figure

Figure 9: Kaplan-Meier Survival Curves by Parcel Characteristic



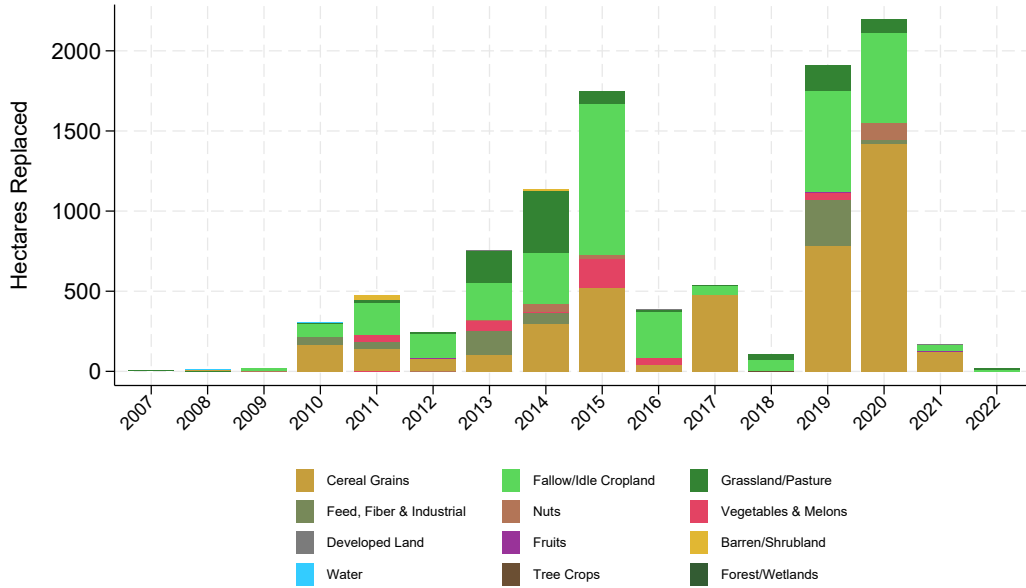
Notes: Each panel compares Kaplan-Meier survival curves for parcels above and below the median of the indicated time-invariant parcel characteristic.

A9 contextualizes the magnitude of solar developments shown in Figure 10 by plotting land uses as shares of the fixed total area of never-urban land in the Central Valley. In this figure, areas converted to solar collapse into the nearly invisible yellow bar-segments, emphasizing that solar additions are minuscule relative to total land area.

5.2 Main Regressions

Table 1 presents our main results, quantifying the parcel-level determinants of solar adoption. The mean adoption probability in our sample is 0.014%. Our baseline LPM specification (column 1) confirms that all hypothesized factors significantly predict solar development. A one-standard-deviation (SD) increase in drought exposure increases the probability of solar adoption by 0.0145 percentage points (pp), an effect roughly equivalent in magnitude to the baseline adoption probability. A one-SD increase in log farm labor costs raises adoption probability by 0.0057pp. A one-SD increase in solar irradiance increases adoption probability by 0.0068pp, while a one-SD increase in log inverse distance to the nearest substation increases adoption probability by 0.0053pp. Proximity to transmission lines has a smaller (though still

Figure 10: Crops Replaced by Solar



Notes: Crop coverage data are from [USDA National Agricultural Statistics Service \(2023\)](#). Crops included in each category are reported in Appendix Table A1. Areas of each crop type converted to solar are calculated based on crop cover observed in the year prior to solar installation.

significant) effect (0.0032pp), consistent with substations serving as the primary grid interconnection point. Larger parcels are also significantly more likely to adopt solar: a one-SD increase in log area increases adoption probability by 0.0097pp. Finally, parcels with poor soil quality are 0.0166pp more likely to adopt solar than those with good soil. These results are visualized in Appendix Figure A10, showing that poor soil quality and drought exposure are the strongest predictors of solar adoption in the corrected LPM.

Column (2) reports corresponding estimates from the Cox proportional hazards model. All covariates are statistically significant and point in the same direction as column (1): a one-SD increase in drought exposure, labor costs, and poor soil quality raise the adoption hazard by factors of 1.92, 7.32, and 4.88, respectively, while a one-SD increase in parcel area, substation proximity, and transmission line proximity raise it by factors of 3.08, 1.65, and 1.43 respectively. Solar irradiance exhibits the largest hazard ratio (HR = 22.94). These results are visualized in Appendix Figure A11, alongside corrected logit and cloglog estimates for robustness. Results are relatively stable across models, with solar irradiance, farm labor costs, poor soil quality, and parcel area emerging as the strongest predictors of solar adop-

tion. Appendix Table A4 reports LPM and Cox estimates in natural (unstandardized) units for reference.

Table 1: Determinants of Solar Adoption

	Corrected LPM (per 10k)	Cox (HR)
Drought Exposure (t-1 to t-3)	1.4471*** (0.2140)	1.9214** (0.5961)
Labor Cost (t-1 to t-3, log)	0.5713*** (0.0597)	7.3168*** (1.2244)
Poor Soil Quality	1.6602*** (0.2201)	4.8829*** (1.0054)
Solar Irradiance	0.6801*** (0.0604)	22.9377*** (4.8662)
Proximity to Substation	0.5302*** (0.0920)	1.6476*** (0.1190)
Prox. to Transmission Lines	0.3247*** (0.0656)	1.4279*** (0.0815)
Parcel Area (log)	0.9652*** (0.0702)	3.0814*** (0.2519)
Year Fixed Effects	Yes	No
Observations	3,016,944	3,016,944

Notes: Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Column (1) reports coefficient estimates with standard errors in parentheses for a linear probability model with year fixed effects. The dependent variable is rescaled to solar adoptions per 10,000 parcel-years at risk, such that a coefficient of 1.0 indicates one additional adoption per 10,000 parcel-years for a one-standard-deviation increase in the covariate. Column (2) reports hazard ratios from a Cox proportional hazards model with nonparametric baseline hazard. Values > 1 indicate higher adoption hazard. All continuous covariates are standardized to mean zero, unit SD, and standard errors are clustered at the parcel level.

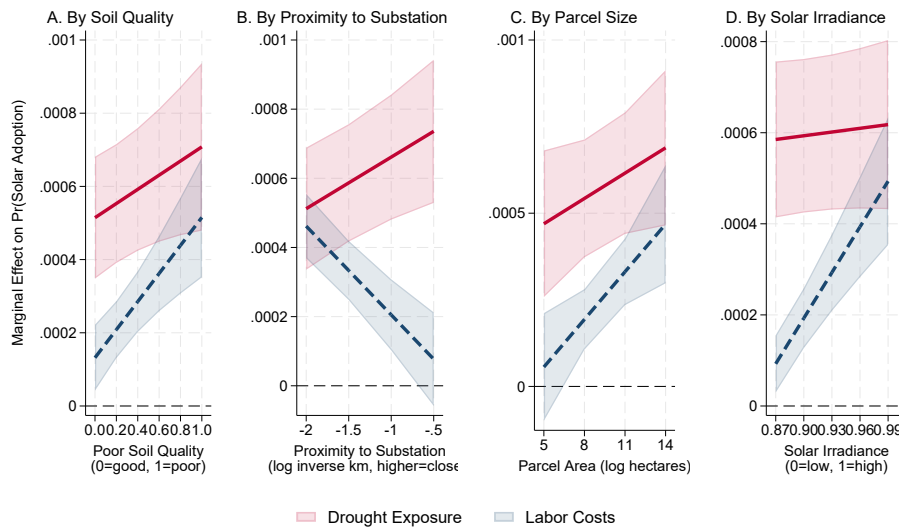
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Interaction Effects

Appendix Table A5 examines whether drought's effect on solar adoption interacts with other parcel characteristics. Corrected LPM estimates, reported in Column (1), reveal that drought's main effect remains positive and significant even after including interactions, and this effect is significantly larger on parcels with poor soil quality, closer proximity to a substation, and greater parcel area. Column (3) reports corresponding Cox estimates: drought's main effect is greater than 1 and significant ($HR = 3.0423$), but the only significant interaction is with solar irradiance ($HR = 0.7275 < 1$), indicating that drought's hazard ratio shrinks toward 1 as solar irradiance increases. Columns (2) and (4) report analogous estimates for corrected logit and cloglog models, with results discussed in Section 5.9 (Robustness). To explore interactions in more detail, Figure 11 plots marginal effects of lagged drought exposure and

farm labor costs on solar adoption across fixed parcel characteristics, using corrected LPM estimates. Drought and labor cost marginal effects are positive and increase with worsening soil quality (11a), parcel size (11c), and solar irradiance (11d), consistent with drought and labor cost pressures mattering most where solar development is otherwise more feasible and attractive. Figure 11b indicates that proximity to substations strengthens drought’s marginal effect on solar adoption but weakens the marginal effect of labor costs – which is nevertheless positive along the entire parcel size distribution.

Figure 11: Marginal Effects of Drought Exposure and Labor Costs by Parcel Characteristic



Notes: Figure plots marginal effects of lagged drought exposure and farm labor costs on solar adoption probability across parcel characteristics, estimated from a linear probability model with both drought and labor cost interactions. Panel A varies soil quality from 0 (excellent agricultural suitability) to 1 (poor agricultural suitability). Panel B varies inverse proximity to electrical substations, measured as the normalized log inverse distance. Panel C varies log parcel area. Panel D varies normalized solar irradiance. Values excluded from these ranges account for less than 1% of all parcels and no solar parcels. Shaded areas represent 95% confidence intervals based on robust standard errors clustered at the parcel level. All other variables are held at sample means.

5.4 Intensive Margin: Scale of Adoption

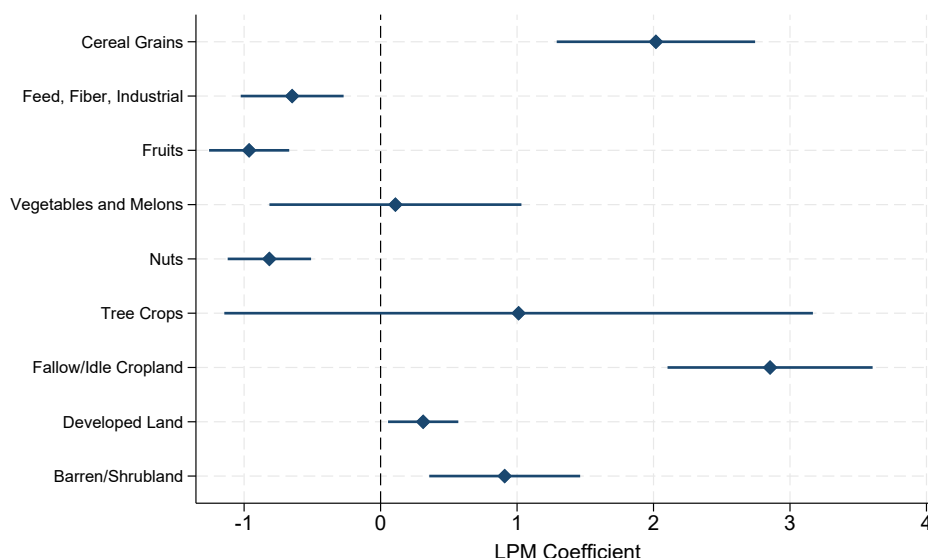
We next examine the intensive margin of adoption to assess determinants of installation size. Appendix Table A6 restricts the sample to adopting parcels and regresses installed solar area (in hectares) on standardized covariates using OLS. In this (much smaller) sample of parcels that ever adopt solar, drought exposure is positive but no longer statistically signifi-

cant, suggesting that while drought predicts *whether* a parcel adopts solar, it does not predict *how much* land is converted once a parcel owner has decided to adopt. Scale of adoption is primarily driven by labor costs (+7.51 ha per one-SD increase), solar irradiance (+8.43 ha), proximity to substation (+2.61 ha), and parcel area (+8.92 ha – the strongest single predictor, mechanically). Drought’s role re-emerges through significant positive interactions with solar irradiance, substation proximity, and parcel area, indicating that drought exposure amplifies the scale of solar development specifically where site conditions already favor larger installations. These results are visualized in Appendix Figure A12.

5.5 Prior Land Use

Certain land uses are substantially more valuable than others and may differ in their suitability for conversion to solar. We re-estimate the corrected LPM including categorical variables for each land use type in period $t - 1$, and report coefficient estimates in Figure 12. Full results are reported in Appendix Table A7. Figure 12 reveals substantial heterogeneity in conversion propensity across land types. Relative to the reference category (grassland/pasture), fallow and idle cropland is most likely to be converted to solar ($\hat{\beta} = 2.8542$, equivalent to a 0.0285pp increase in adoption probability, or approximately twice the sample mean adoption rate), followed by cereal grains ($\hat{\beta} = 2.0175$), barren/shrubland ($\hat{\beta} = 0.9094$), and developed open space ($\hat{\beta} = 0.3121$). These categories represent relatively flexible or low-value uses that can be readily repurposed for utility-scale solar without significant clearing or land preparation. In contrast, high-value crops show markedly lower conversion rates: nuts ($\hat{\beta} = -0.8149$), fruits ($\hat{\beta} = -0.9633$) (both often perennials), and feed/fiber/industrial crops, e.g., flaxseed, cotton, sunflower ($\hat{\beta} = -0.6479$) are all significantly less likely to be converted than grassland/pasture. Tree crops represent a niche category (e.g., Christmas trees) and are imprecisely estimated. These patterns demonstrate that solar development does not randomly displace agricultural land, but instead concentrates on parcels with lower agricultural opportunity costs, such as those lying fallow, dedicated to low-value annual crops, or supporting livestock grazing. Appendix Figure A13 plots analogous results from corrected logit, Cox, and Cloglog models for robustness. Estimates are stable across models, confirming that fallow/idle cropland and cereal grains are the most likely land uses to be converted to solar, while nuts and fruits are the least likely.

Figure 12: Effect of Prior Land Use on Solar Adoption



Notes: Figure plots coefficient estimates and 95% confidence intervals from linear probability model showing the effect of prior land use (t-1) on solar adoption probability. The omitted reference land use is grassland/pasture. Water and forest/wetland parcels are excluded due to high imprecision in estimates due to low frequency in the sample. Estimates for each of these categories are not statistically significant.

5.6 Groundwater Management, Irrigation, and Fallowing

Access to groundwater for irrigation is another critical determinant of agricultural viability in the Central Valley. California's Sustainable Groundwater Management Act (SGMA), enacted in 2014, requires groundwater basins statewide to bring pumped water supply and demand into balance by the early 2040s (Hanak et al. 2019). Under the SGMA, local Groundwater Sustainability Agencies were required to form by 2017 and submit Groundwater Sustainability Plans by 2020 or 2022 depending on each basin's priority designation. Basins that are critically overdrafted face the most stringent pumping restrictions (Hanak et al. 2019).¹⁰ Groundwater pumping reductions under SGMA are expected to reduce water availability for agriculture in the Central Valley by 20%, requiring fallowing of up to 900,000 acres of irrigated land by 2040 (Escriva-Bou et al. 2023). Groundwater recharge and pumping rates are

¹⁰SGMA implementation has been gradual. Groundwater Sustainability Agencies were still forming and submitting plans for state review through the early 2020s, and the first state-enforced pumping restrictions did not begin until 2024, after the end of our sample period. As a result, SGMA did not impose binding legal constraints on pumping during our study, though its passage may have shaped landholders' expectations about future water availability and, in turn, the relative attractiveness of solar adoption.

themselves sensitive to drought: during dry years, farmers pump more heavily from aquifers to compensate for reduced surface water availability, while recharge slows as less water percolates into the ground. We find a weak positive correlation (0.189) between average drought exposure and groundwater well depth across parcels over 2006–2023 (Appendix Figure A14).

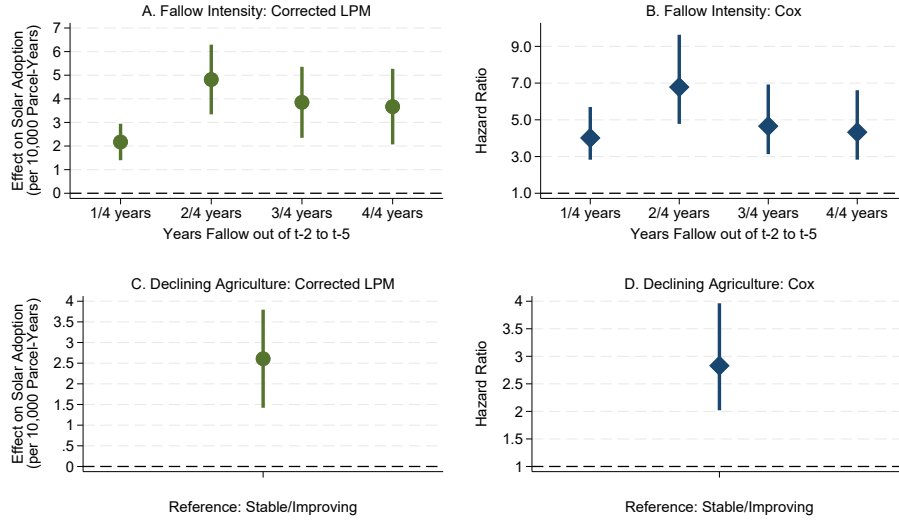
Groundwater supplies approximately two-thirds of water for irrigation during drought in the Central Valley, and groundwater levels have been declining at an accelerating rate since 2003 (Liu et al. 2022). We examine impacts of groundwater depletion on solar adoption using an inverse-distance-weighted measure of well depth for each parcel (measured from the nearest test well available in the SGMA Groundwater database), which serves as a proxy for the difficulty of accessing groundwater. When included alongside our standard covariates (Appendix Table A8), well depth is significantly positively associated with solar adoption in both the corrected LPM (1.2171) and Cox (HR = 1.5120) specifications. In the LPM, drought exposure remains positive and significant (1.1083) even after including well depth, indicating that drought and groundwater access each independently predict solar adoption, rather than one channel proxying for the other. In Cox, logit, and cloglog specifications (Appendix Table A9), the drought effect remains positive but loses statistical significance when well depth is included, suggesting that in these specifications, groundwater stress captures some of the effect previously captured by drought stress. Coefficients on other covariates remain similar in sign and significance to our main results in Table 1.

Land fallowing is a downstream result of drought and groundwater stress. To assess whether solar serves as an adaptation strategy for lands with increasing propensity for fallowing, we measure the number of years each parcel was fallow or idle out of the four years prior to adoption ($t - 2$ to $t - 5$, omitting $t - 1$ to avoid fallowing resulting mechanically from solar site preparation and construction), and separately classify each parcel’s recent agricultural trajectory based on whether its fallow status changed within that window.¹¹ Figure 13 reports estimates from both corrected LPM and Cox models.

Sub-figures 13a and 13b indicate that solar adoption probability increases with fallow intensity, but not monotonically: the corrected LPM effect rises from $\hat{\beta} = 2.2$ per 10,000 parcel-

¹¹Parcels that were cropped in $t - 4$ and $t - 5$ but became fallow in $t - 2$ or $t - 3$ are classified as deteriorating; all other parcels – including those with consistently fallow or consistently cropped status throughout the window, as well as those recovering from fallow to cropping – constitute the reference category. Both intensity and trajectory specifications control for drought exposure, labor costs, and other parcel characteristics.

Figure 13: Effect of Fallow Intensity on Solar Adoption



Note: Top panels show the effect of fallow intensity, measured as the number of years out of $t - 2$ through $t - 5$ classified as fallow/idle cropland, on solar adoption, relative to a reference category of 0/4 years fallow. Bottom panels show the effect of deteriorating agricultural trajectory relative to stable or improving parcels. Deteriorating parcels were actively cropped at $t - 4$ and $t - 5$ but fallow by $t - 2$ or $t - 3$. Improving parcels show the reverse transition, from fallow at $t - 4$ or $t - 5$ to cropped by $t - 2$ and $t - 3$, while stable parcels include those with an unchanged fallow status (either consistently cropped or consistently fallow) throughout $t - 2$ to $t - 5$. All measures omit $t - 1$ to avoid mechanical fallowing during solar site preparation. Left panels report linear probability model coefficients (outcome: solar adoptions per 10,000 parcel-years). Right panels report hazard ratios from Cox proportional hazards models. All models include controls for lagged drought exposure and agricultural labor costs, soil quality, solar irradiance, proximity to electrical substations, and parcel size. Linear probability models additionally include year fixed effects. All continuous controls are standardized to mean zero, unit SD. Post-adoption observations are excluded (corrected risk set). Standard errors are clustered at the parcel level.

years for parcels that were fallow in just one of the four years preceding adoption, to 4.8 for parcels fallow in two of the four years, to 3.9 and 3.7 for parcels fallow in three and four years, respectively. Cox estimates follow the same pattern, with hazard ratios rising from 4.0 (one year fallow) to 6.9 (two years fallow), 4.6 (three years fallow) and 4.3 (four years fallow). All estimates are statistically significant. These results suggest that fallowing land for 2+ years out of four may be a strong signal to parcel-owners to adopt solar. Sub-figures 13c and 13d indicate that parcels transitioning from cultivation to fallow (deteriorating) are significantly more likely to adopt solar than those that are stable or returning to cultivation (improving).

Collectively, these results indicate that solar adoption concentrates on parcels facing compounding forms of agricultural stress – groundwater scarcity, fallowing intensity, and recent

shifts toward fallowing – consistent with our broader finding that solar development serves as an adaptive response to declining agricultural viability.

5.7 Prime, Unique, and Important Farmland

The U.S. Department of Agriculture classifies agricultural land into several quality tiers to inform land use decisions. *Prime farmland* has the best combination of physical and chemical characteristics for producing food, feed, forage, fiber, and oilseed crops, including adequate and dependable water supply, favorable temperature and growing season, and minimal erosion or flooding risk. *Unique farmland* does not meet Prime farmland criteria, but is used to produce specific high-value crops such as citrus, tree nuts, olives, and other fruits and vegetables, reflecting a special combination of soil and microclimate conditions. *Important farmland* is a broader category covering lands of statewide or local importance or that are otherwise designated for agricultural production by state or local agencies, without meeting stricter Prime or Unique classifications (USDA 2026). Winikoff et al. (2026) study these classifications' association with solar development on agricultural parcels nationally. They find little correlation between Prime classification and solar adoption, but a significant positive correlation between Important classification and solar, suggesting that solar is not disproportionately installed on the highest quality farmland, but is often sited on lands considered important for agriculture.

We re-estimate our baseline specification including indicators for Prime, Unique, and Important farmland (Appendix Table A10). Consistent with Winikoff et al. (2026), we find no evidence that solar development disproportionately targets the highest-quality agricultural land in the Central Valley. In fact, parcels designated as Prime farmland by the USDA are significantly less likely to adopt solar in the corrected LPM ($\hat{\beta} = -1.6864$) and shows a hazard ratio well below one in the Cox model (HR = 0.2925). As reported in Appendix Table A11, this pattern holds across corrected logit (AME = -0.0001) and Cloglog models ($\hat{\beta}^{clog} -1.3578$) as well. Unique farmland shows the same significantly negative pattern across all four estimators (LPM: -1.0720; Cox HR: 0.4359; logit: -0.0001; cloglog: -0.9240), consistent with landholders preserving the highest value and most specialized agricultural lands rather than converting them to solar. Important farmland shows a more mixed pattern: the corrected LPM effect is small and statistically insignificant ($\hat{\beta} = -0.1244$), but the Cox, logit, and cloglog specifications all show a significant positive association (Cox HR: 1.5035; logit AME: 0.00001; Cloglog:

0.3582), echoing [Winikoff et al. \(2026\)](#)’s finding that solar development is more likely on land that is not Prime or Unique but is designated as Important by state or local agencies.

5.8 Neighbor Effects (Spatial Spillovers)

Solar adoption may also depend on decisions made by nearby landholders. Two mechanisms could generate such spatial spillovers. First, peer learning: neighboring solar installations may reduce uncertainty about permitting timelines, construction costs, and lease terms, making adoption more attractive once a nearby landholder has already navigated the process. Second, local scale economies: parcels near existing solar facilities may benefit from shared grid interconnection infrastructure, established relationships with developers and contractors, or lower marginal costs of extending transmission access to an already-served area. Both mechanisms predict that solar adoption should cluster spatially beyond what is explained by parcel characteristics and drought or labor cost exposure alone.

We test for spatial spillovers by adding measures of neighboring solar development, lagged one year to avoid simultaneity with the adoption decision, to our baseline specification (Table [A12](#)). Using an indicator for whether any solar facility existed within a given radius around a parcel, we find a significant, positive effect at 2km ($\hat{\beta} = 0.6053$) and 5km ($\hat{\beta} = 0.3021$), as well as a positive but statistically insignificant effect at 10km ($\hat{\beta} = 0.0970$), suggesting spillovers decay over distance and operate at a relatively local scale. Using the count of solar facilities within each radius instead of a binary indicator, the pattern is similar: an additional solar facility installed as of $t - 1$ within 2km of a parcel significantly increases that parcel’s likelihood of solar adoption in t by 0.0382, with the effect decaying to 0.0225 within 5km and 0.0174 within 10km. Coefficients on drought exposure, labor costs, and other main effects remain stable in magnitude and significance across all six specifications relative to Table [1](#), indicating that neighbor effects operate alongside, rather than in place of, the parcel-level determinants of adoption identified in our main results.

5.9 Robustness

We conduct numerous robustness checks to assess the sensitivity and consistency of our findings. First, we confirm that our results are robust to alternative panel regression and sur-

vival model estimators: Appendix Table A13 reports corrected logit and cloglog estimates to complement the preferred LPM and Cox specifications, and Appendix Figure A11 plots effects across corrected logit, Cox, and cloglog estimators, showing consistent signs and significance for most covariates. Cloglog estimates for farm labor costs and solar irradiance are lower than for other models, but remain positive and significant, while the cloglog estimate for drought remains positive but is not statistically significant. Appendix Figure A13 plots land-use coefficients across corrected logit, Cox, and cloglog estimators, showing that estimates are highly consistent and stable. Appendix Table A5 reports interactions for cloglog alongside corrected LPM and Cox, with drought interaction terms losing significance in this specification. Effects of well depth and Prime/Unique/Important farmland classifications on solar adoption are strongly consistent across all estimators (Appendix Tables A9 and A11).

Second, we re-estimate corrected LPM and Cox proportional hazards specifications using alternative lag structures for drought exposure and agricultural labor costs, replacing our preferred 3-year lag with 1-year and 5-year lags. Separately, we re-estimate main specifications omitting the poor soil quality measure, which is unavailable for parts of the Central Valley and mechanically excludes those areas, including 27 solar facilities, from our main sample (Appendix Table A2). Labor cost effects are stable in sign and significance across all three specifications. Drought exposure effects are substantially larger under the 5-year lag (LPM: 2.1672; Cox HR: 4.6678) relative to our preferred 3-year specification, consistent with landholders responding more strongly as drought exposure accumulates over time. Under the 1-year lag, drought's effect is weakly negative in the corrected LPM ($\hat{\beta} = -0.4924$) and statistically indistinguishable from zero in the Cox model, indicating that parcel-owners do not respond immediately to drought by adopting solar. Omitting soil quality does not change the sign or significance of estimates for remaining covariates.

Third, to control for time-invariant unobservables at the county and parcel level – including concerns that our results reflect broader urbanization or development pressure rather than agricultural opportunity-cost mechanisms – we include county and parcel fixed effects (Appendix Table A14). County fixed effects absorb time-invariant county-level differences in urbanization, land values, policies or ordinances, and development pressure, identifying labor cost effects from within-county variation over time. After including county fixed effects, farm labor costs remain positive and significant (LPM $\hat{\beta} = 0.9330$; Cox HR: 17.9216), con-

firming these results are not driven by cross-county differences correlated with urbanization. Parcel fixed effects absorb time-invariant parcel-level unobservables, including owner characteristics. After including these fixed effects, farm labor costs remain positive and significant (LPM $\hat{\beta}$: 1.2551; Cox does not accommodate parcel fixed effects). Under each of these types of fixed effect, drought exposure becomes statistically insignificant, suggesting that drought affects solar adoption over a time horizon that is slow enough to be mostly absorbed by time-invariant parcel or regional components. In the specification with parcel fixed effects and drought and labor interactions, only the drought $\times$ substation proximity interaction is positive and significant, while labor cost interactions with poor soil quality, solar irradiance, substation proximity, and parcel area are all positive and significant. As a more targeted test of urbanization pressure as a potential confounder, we replace county fixed effects with Rural-Urban Continuum Code (RUCC) fixed effects based on each county's baseline (2003) classification, along with a dummy control for whether a county became more urban between 2003 and 2023 (Appendix Table A15). Labor costs, poor soil quality, solar irradiance, parcel area, and both grid proximity measures remain significant and stable across all four estimators. Drought exposure remains positive and stable in magnitude across estimators, but is only significant in the corrected LPM. Among the RUCC coefficients themselves, the largest is for metro areas with under 250K population (relative to both more urban and less urban counties), which is a significant predictor of solar development in the LPM (though not in other models). This suggests solar concentrates in moderately urban areas, balancing proximity to demand centers against the higher land costs and congestion of more urbanized areas.

Fourth, we conduct a falsification test to distinguish farm-specific labor cost pressure from generalized urbanization or development pressure by replacing the farm wage in our preferred specification with the deflated all-sector average county wage (Appendix Table A16).¹² If our results were driven by generic urbanization pressure rather than agricultural labor costs specifically, the all-sector wage should perform similarly to farm wages. Instead, the all-sector wage is significantly *negatively* associated with adoption in the LPM ($\hat{\beta} = -0.3554$) and corre-

¹²Urbanization represents a substantially larger threat to Central Valley farmland than solar development. Between 2006 and 2023, 131,275 parcels (including 77,889 previously cultivated parcels) in the Central Valley were converted to urban use, relative to 419 that were converted to utility-scale solar installations. In other words, 186 times more agricultural parcels were lost to urban expansion than were lost to solar during this period. Observing that solar installations are never built on urban parcels during the study period, we drop all ever-urban parcels to retain a strongly balanced panel of never-urban parcels for our analysis.

sponds to a hazard ratio below one in the Cox model ($HR = 0.6412$) – the opposite sign from our preferred farm-wage specification. This falsification result supports our interpretation that it is specifically declining agricultural profitability, not broader development pressure, that predicts solar adoption.

Fifth, we re-estimate the prior land-use analysis with land use measured at $t - 2$ rather than $t - 1$, since land use at $t - 1$ may already reflect solar site preparation or construction activity (Appendix Table A17 and Figure A15). Results are broadly consistent. Fallow/idle cropland, cereal grains, barren/shrubland, and developed land remain positive and statistically significant, while fruits and nuts remain negative and significant. Vegetables and melons shifts from insignificant to positive and significant, while feed, fiber, and industrial crops shifts from negative and significant to statistically insignificant.

Finally, we address potential reverse causality in our 2022 measure of electrical substation locations, which could have been built in response to solar siting. To do so, we use electricity infrastructure data from 2014, obtained through a data request to the California Energy Commission, which capture substation locations prior to the bulk of utility-scale solar development in our sample. Due to less complete reporting in the 2014 data, we treat this as a robustness check rather than our main proximity definition. We re-estimate our main specifications using proximity to 2014 substations, rather than 2022 substations, with results reported in Appendix Tables A18 and A19. Coefficient estimates for the proximity to substation covariate are stable across earlier and later measurements: $\hat{\beta} = 0.5302^{***}$ (2022) vs 0.6281^{***} (2014) for LPM and $HR = 1.6476^{***}$ (2022) vs 1.3092^{***} (2014) for Cox, suggesting that substations are, for the most part, constructed exogenously to solar siting.

5.10 Priority Solar Siting

To prioritize future utility-scale solar development in the Central Valley, we construct a continuous solar suitability index that identifies high-potential parcels that had no solar installation as of 2023. The index applies linear reward and penalty functions over six dimensions: proximity to electric substations, parcel size, drought exposure, solar irradiance, soil quality, and low-value land-use, which are aggregated into a weighted composite based on standardized coefficient estimates from Appendix Figure A10 and normalized to a 0-100

scale.¹³ The distribution of index values is shown in Appendix Figure A16. Figure 14 maps this index for the Central Valley, revealing clusters of parcels with high solar potential and low agricultural potential – particularly along the Valley’s northwest edge and around Chico and Sacramento.¹⁴ As a conservative targeting rule, we identify high-suitability parcels as those scoring above the 99.9th percentile of the continuous solar suitability index. Appendix Figure A19 maps the resulting 328 parcels, totaling 100,967ha, which could support approximately 25.2GW of generating capacity, or 43.8% of California’s 2045 utility-scale solar energy target (Weaver 2024; CA Public Utilities Commission 2024), while concentrating development on marginal agricultural land with strong solar feasibility in the Central Valley alone.¹⁵

To assess predictive power of the suitability index, we conduct an out-of-sample validation exercise. We estimate the model using only data from 2006-2018 and apply the resulting standardized coefficients to 2018 parcel characteristics to generate predicted suitability scores. We then divide all non-solar parcels as of 2018 into deciles based on these predicted scores and examine whether actual solar installations occurring between 2019-2023 concentrate in higher-predicted-suitability deciles. Appendix Figure A20 shows strong out-of-sample predictive power: post-2018 solar adoption rates increase from near-zero in the bottom deciles to 0.12-0.14% in each of deciles 8-10.¹⁶ Notably however, 41% of post-2018 installations (61 of 147) occurred outside the top two suitability deciles, with 13% occurring outside the top three deciles. While this could reflect unobserved factors such as landholder preferences, it also suggests scope for siting policies that guide development toward parcels with high solar

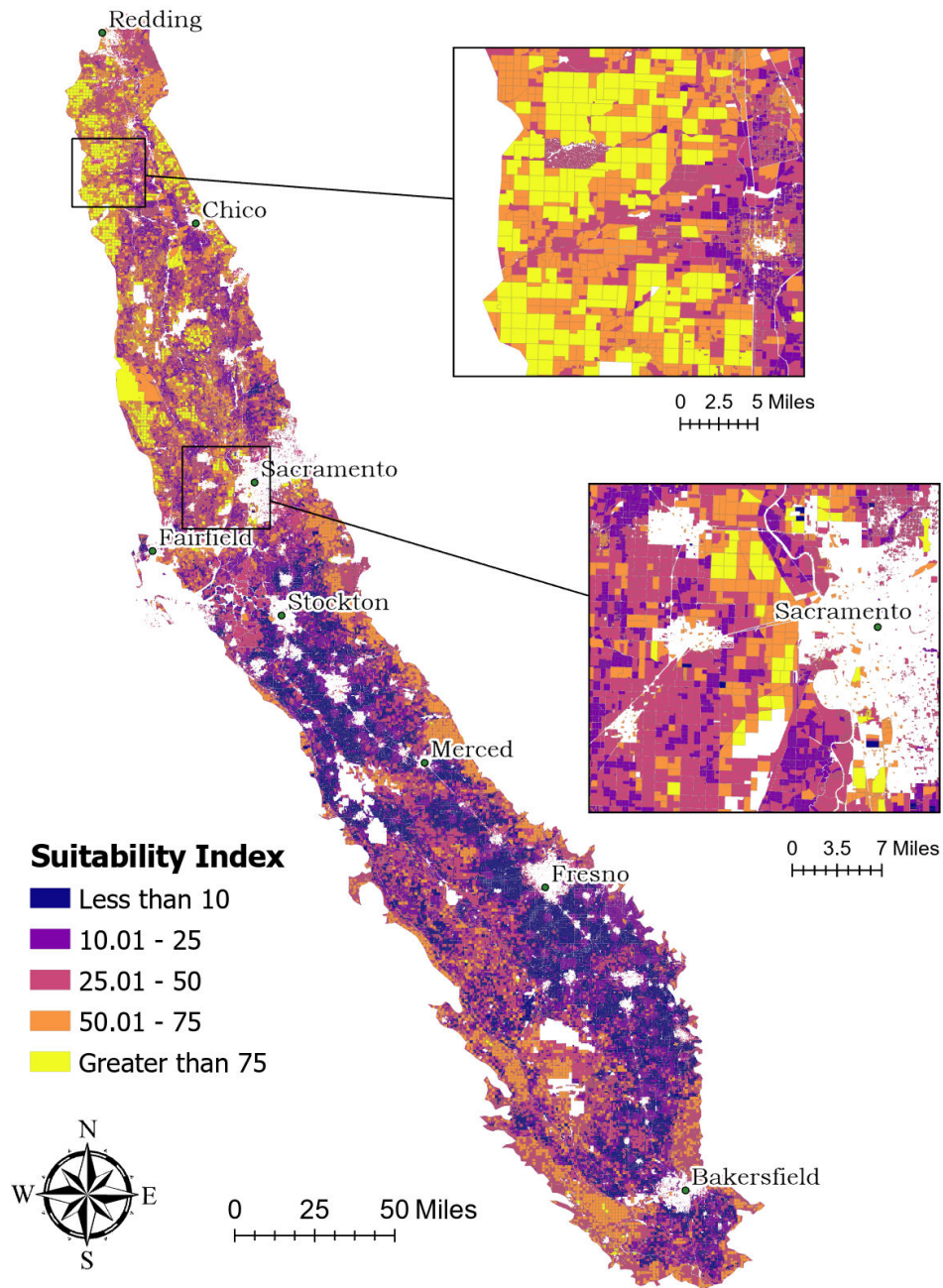
¹³Substation proximity is scored at full value at 0km and declines linearly to zero at 4km. Parcel size is scored at zero below 40ha and increases linearly to full value at 200ha. For drought exposure, solar irradiance, and soil quality, we assign scores based on empirical distributions within the study region. Drought exposure receives zero below the 60th percentile, increases linearly from the 60th to the 80th percentile, and reaches full value above the 80th percentile. Solar irradiance receives zero below the 20th percentile, increases linearly from the 20th to 40th percentile, and reaches full value above the 40th percentile, reflecting generally high GHI across the region. Soil quality is scored inversely: parcels receive zero above the 40th percentile, increase linearly from the 40th to the 20th percentile, and receive full value below the 20th percentile. Low-value land uses (defined as grassland/pasture, fallow/idle cropland, developed open space, and barren/scrubland), receive full value; all other land uses receive zero. To incorporate this sixth factor, we re-estimate the standardized coefficient model including the low-value land use indicator, yielding empirically-derived weights for all six components.

¹⁴Appendix Figure A17 presents an alternative simple count index using binary versions of the same criteria. Appendix Figure A18 confirms a strong positive correlation between the two measures.

¹⁵Megawatt values are calculated based on an assumption of 0.5MW per hectare of solar panels and 50% of parcel area dedicated to panels, which is the sample average.

¹⁶Appendix Figure A21 confirms this pattern by plotting marginal effects from a logistic regression of post-2018 solar adoption on predicted solar suitability deciles as of 2018, revealing a significant increase in adoption likelihood in the top three deciles. Appendix Figure A22 plots the ROC curve with an area under the curve of 0.803, indicating that the suitability index correctly ranks parcels by likelihood of solar adoption 80% of the time.

Figure 14: Continuous Solar Suitability Index



Notes: Map shows the continuous solar suitability index (0-100) for non-urban parcels in California's Central Valley with no solar installation as of 2023. The index combines weighted penalty functions for six characteristics: proximity to electrical substations (full value at 0km, declining linearly to zero at 4km), parcel size (zero below 40ha, full value above 200ha), drought exposure (rewarded above 60th percentile, full value above 80th percentile), solar irradiance (rewarded above 20th percentile, full value above 40th percentile), poor soil quality (rewarded below 40th percentile, full value below 20th percentile), and low-value land use (full value for fallow/idle cropland, grassland/pasture, developed open space, or barren/shrubland; zero otherwise). Weights are based on standardized regression coefficients. Parcels with low/medium/high-density development are set to zero.

potential and low agricultural opportunity costs.

Finally, as a policy-oriented alternative prioritization, we construct a second suitability index focused on concerns raised by introduction of the Sustainable Groundwater Management Act and USDA’s prohibition of loans or subsidies for solar panels installed on Prime farmland ([USDA 2025c](#)). This index identifies parcels if they are not classified as Prime, Unique, or Important, have well depth above the 90th percentile (indicating vulnerability to SGMA-driven fallowing), are not classified as developed, water, wetland, or forest, and are otherwise solar-suitable (i.e., >40ha, <2km of a substation, >75th percentile of solar irradiance). Appendix Figures [A23–A24](#) report the results, revealing a distinct pattern with high-priority parcels concentrated in the southern extreme of the Central Valley. This policy-oriented index identifies 135 parcels meeting all six criteria, which could support 3.3GW of solar capacity. Parcels that meet at least five of six of the criteria would expand this potential solar capacity to 46.3GW.¹⁷

6 Discussion

California’s Central Valley is an ideal setting to study land-use tradeoffs arising from the transition to renewable energy. One of the world’s most productive agricultural regions, the Central Valley supplies a quarter of U.S. food production, yet faces mounting pressure from drought and groundwater depletion. Simultaneously, California’s emission reduction targets aim for 57.5 gigawatts of new utility-scale solar capacity by 2045, much of which will compete for the same lands that currently support agriculture. The central question is whether solar development threatens agricultural communities or offers an economic lifeline amid changing environmental conditions.

Our findings reveal systematic patterns in utility-scale solar adoption. Consistent with a conceptual framework where landholders choose between agriculture and solar to maximize expected returns from their land based on parcel characteristics and time-varying agricultural profitability shocks, we find that solar adoption concentrates on parcels where agriculture is least viable and solar development is most attractive. Solar adoption increases significantly in association with parcel-level drought exposure, agricultural labor costs, poor soil quality,

¹⁷Our replication package, available at [Replication Package Link TBD], allows stakeholders to reconstruct or customize these prioritization criteria.

following frequency, groundwater depth, solar irradiance, proximity to electrical substations, and parcel size.¹⁸ High-value permanent crops such as fruits and nuts are rarely converted, while low-value land uses such as fallow/idle cropland, grassland/pasture, and barren land are disproportionately selected. Applying these criteria to prioritize parcels where solar benefits are high and agricultural opportunity costs are low, we identify 328 optimal sites that could generate 25.2 GW – representing 44% of California’s 2045 utility-scale solar target – while concentrating development on agriculturally marginal lands.

This study has several limitations. First, we do not observe landholder characteristics, which could shape solar adoption in ways our parcel-level covariates cannot capture. Second, our framework focuses on individual land-use decisions and does not account for broader community impacts, including effects on agricultural service sectors, local tax revenues, or landscape aesthetics. Solar installations could potentially impose negative externalities, such as visual impacts, and positive externalities, such as reduced competition for scarce groundwater. Third, we only observe solar installations exceeding 1 MW, potentially understating total solar deployment and missing distinct siting patterns for smaller projects. Finally, our analysis is limited to California’s Central Valley, which is in many ways a unique agricultural region – raising concerns over external validity. The core economic pressures we study, however, are not unique to California. Drought reduces crop yields and farm income nationwide ([Kuwayama et al. 2019](#)), and over 40% of the United States experienced significant drought between 2020-2023 ([Holmes and Kuwayama 2026](#)). Similarly, farm labor costs are high nationwide: real farm wages have grown faster over the past decade than in the preceding twenty years, a trend the USDA attributes to farmers’ difficulty finding workers ([USDA 2025a](#)). Soil quality, groundwater availability for irrigation, and other factors we study are common concerns for farmers everywhere, suggesting patterns we document in the Central Valley likely generalize beyond California.

Our findings challenge the view that expansion of utility-scale solar necessarily threatens agricultural communities. Solar adoption concentrates precisely where farming faces the greatest challenges – on drought-exposed, high-cost parcels with poor soil and limited groundwater – in line with benefit-maximizing decisions by landholders. This pattern sug-

¹⁸While nearly all covariates are highly stable in sign, magnitude, and significance across models, fixed effects, interactions, and additional controls, drought exposure effects are sensitive to model choice and the inclusion of county and parcel fixed effects, and are best read as suggestive rather than definitive.

gests that strategic siting policies can reconcile competing imperatives to expand renewable energy generation while preserving agricultural production. More broadly, our results reveal solar leasing as a potential adaptation mechanism that may help sustain rural economies as climate change intensifies, providing stable returns when agricultural viability declines.

Declaration of Generative AI Use

During the preparation of this work, the authors used Claude (Anthropic) to assist with code debugging, text revision, and proofreading. The authors reviewed and verified all outputs and take full responsibility for the content of this publication. All substantive content, research design, and analysis are the authors' original work.

Declaration of Competing Interests

The authors declare no competing interests.

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ONLINE APPENDIX

Contents

Appendix [A](#): Supplementary Tables

Appendix [B](#): Supplementary Figures

A Supplementary Tables

Table A1: Reclassification of Crop Categories

Aggregated Classification	Original Raw Classification
Cereal Grains	Corn, Rice, Sorghum, Soybeans, Barley, Durum Wheat, Spring Wheat, Other Small Grains, Rye, Oats, Milet, Speltz, Buckwheat, Lentils, Triticale, Dbl Crop Win-Wht/Soybeans, Dbl Crop WinWht/Corn, Dbl Crop Oats/Corn, Dbl Crop Barley/Sorghum, Dbl Crop WinWht/Sorghum, Dbl Crop Barley/Corn, Dbl Crop Win-Wht/Cotton
Feed, Fiber, and Industrial Crops	Flaxseed, Cotton, Sunflower, Canola, Safflower, Alfalfa, Other Hay/ Non Alfalfa, Camelina, Sugarcane, Hops, Clover/Wildflowers, Sod/Grass Seed, Tobacco, Dry Beans, Other Crops
Fruits	Mint, Caneberries, Cherries, Peaches, Apples, Grapes, Citrus, Pears, Prunes, Oranges, Pomegranates, Nectarines, Plums, Strawberries, Apricots, Blueberries, Cranberries
Vegetables and Melons	Mustard, Sweet Corn, Pop or Orn Corn, Sugarbeets, Potatoes, Sweet Potatoes, Misc Veggies and Fruits, Watermelons, Onions, Cucumbers, Chick Peas, Peas, Tomatoes, Herbs, Carrots, Asparagus, Garlic, Cantaloupes, Olives, Honeydew Melons, Broccoli, Peppers, Greens, Squash, Vetch, Lettuce, Pumpkins, Cabbage, Cauliflower, Celery, Radishes, Turnips
Nuts	Peanuts, Almonds, Walnuts, Pecans, Pistachios
Tree Crops	Christmas Trees, Other Tree Crops
Fallow/ Idle Cropland	Fallow/ Idle Cropland
Grassland/ Pasture	Switchgrass, Grassland/ Pasture
Water	Aquaculture, Open Water, Perennial Ice/ Snow
Developed Land	Developed/ Open Space, Developed/ Low Intensity, Developed/ Med Intensity, Developed/ High Intensity
Barren/ Shrubland	Shrubland, Barren
Forest/ Wetlands	Forest, Wetlands, Deciduous Forest, Evergreen Forest, Mixed Forest, Woody Wetlands, Herbaceous Wetlands

Table A2: Robustness to Alternative Lag Lengths and Sample Restrictions

	1-Year Lag		5-Year Lag		No Soil Quality	
	LPM (per 10k)	Cox (HR)	LPM (per 10k)	Cox (HR)	LPM (per 10k)	Cox (HR)
Drought Exposure (t-1)	-0.4924*	0.9168				
	(0.2944)	(0.1372)				
Labor Cost (log, t-1)	0.4259***	4.9655***				
	(0.0535)	(0.8638)				
Drought Exposure (t-1 to t-5)			2.1672***	4.6678***		
			(0.2238)	(2.1197)		
Labor Cost (log, t-1 to t-5)			0.5500***	5.7431***		
			(0.0543)	(0.9324)		
Drought Exposure (t-1 to t-3)					1.2179***	1.8051**
					(0.1835)	(0.5230)
Labor Cost (t-1 to t-3, log)					0.4378***	6.2163***
					(0.0516)	(0.9890)
Poor Soil Quality	1.5667***	4.6616***	1.8021***	5.1278***		
	(0.2163)	(0.9299)	(0.2266)	(1.0684)		
Solar Irradiance	0.8026***	16.9664***	0.4861***	14.1363***	0.4738***	19.6142***
	(0.0731)	(3.6749)	(0.0447)	(2.7719)	(0.0408)	(4.1680)
Proximity to Substation	0.5162***	1.6362***	0.5278***	1.6555***	0.5262***	1.7256***
	(0.0926)	(0.1192)	(0.0922)	(0.1192)	(0.0853)	(0.1230)
Prox. to Transmission Lines	0.3142***	1.4237***	0.3427***	1.4338***	0.2822***	1.3933***
	(0.0654)	(0.0812)	(0.0663)	(0.0821)	(0.0594)	(0.0786)
Parcel Area (log)	0.9587***	3.1693***	0.9696***	3.1441***	0.9411***	3.3426***
	(0.0701)	(0.2642)	(0.0705)	(0.2591)	(0.0656)	(0.2682)
Drought/Labor Lag	1-Year	1-Year	5-Year	5-Year	3-Year	3-Year
Year Fixed Effects	Yes	No	Yes	No	Yes	No
Observations	2,994,688	2,994,688	3,026,617	3,026,617	3,451,757	3,451,757

Notes: All models restrict the estimation sample to parcel-years up to the year of solar adoption. Columns (1)–(2) use a 1-year lag for drought and labor cost exposure; columns (3)–(4) use a 5-year lag (the preferred specification uses a 3-year average). Columns (5)–(6) omit soil quality, which is unavailable for some Central Valley areas and results in the exclusion of 27 solar facilities from the main sample. Odd columns report coefficient estimates from the corrected LPM with year fixed effects. The dependent variable is rescaled to solar adoptions per 10,000 parcel-years at risk. Even columns report hazard ratios from Cox proportional hazards models. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Crops Replaced by Solar (in hectares), 2007–2022

Year	Cereal Grains	Feed, Fiber & Industrial Crops	Fruits	Vegetables & Melons	Nuts	Tree Crops	Fallow/Idle Cropland	Grassland & Pasture	Water	Developed Land	Barren/Shrubland	Forest/Wetlands
2007	0.82	0.00	0.00	0.00	0.00	0.00	0.00	6.32	0.00	1.26	0.00	0.00
2008	7.33	0.50	1.08	0.56	0.57	0.00	2.51	6.22	0.04	2.13	0.12	0.00
2009	0.77	0.08	1.70	0.00	1.70	0.00	17.03	7.37	0.00	9.86	0.00	0.15
2010	151.73	42.51	0.11	0.16	0.21	0.00	81.15	18.83	0.00	6.67	0.43	0.00
2011	155.27	69.06	7.88	32.75	32.14	0.00	217.35	38.07	0.11	13.21	15.17	0.00
2012	86.59	0.78	7.67	0.00	11.72	0.02	200.53	64.49	0.00	7.20	0.00	0.00
2013	147.96	161.60	6.70	54.18	24.06	0.00	216.04	150.54	0.00	14.35	0.65	1.28
2014	369.81	105.01	50.87	7.83	27.23	0.00	363.23	327.98	0.04	45.20	7.79	1.72
2015	529.56	58.97	11.54	85.44	116.73	0.03	922.76	88.83	1.02	29.50	0.75	0.00
2016	24.07	4.13	0.46	24.20	7.87	0.00	319.17	23.32	0.00	10.94	0.06	0.11
2017	371.75	0.39	0.00	0.00	30.60	0.00	130.99	2.10	0.00	7.25	0.00	0.00
2018	0.03	0.18	0.00	0.00	0.14	2.68	83.18	24.75	0.00	0.80	0.32	0.00
2019	785.95	219.05	9.84	37.56	6.66	0.26	674.37	218.00	29.13	10.79	4.72	5.96
2020	1,341.46	49.98	1.86	28.84	91.17	0.00	605.41	92.99	3.38	26.35	5.60	0.52
2021	126.95	3.14	10.20	0.00	6.41	0.01	23.50	0.52	0.00	1.92	0.03	0.00
2022	0.15	0.05	0.00	0.00	1.17	0.00	5.97	13.04	0.00	0.46	0.00	0.00
Total	4,100.21	715.41	109.91	271.51	358.40	2.99	3,863.17	1,083.38	33.72	187.89	35.65	9.74

Table A4: Results in Natural Units (Unstandardized)

	Corrected LPM (per 10k)	Cox (log-hazard)
Drought Exposure (t-1 to t-3)	5.8135*** (0.8598)	2.6236** (1.2463)
Labor Cost (t-1 to t-3, log)	2.8595*** (0.2988)	9.9609*** (0.8375)
Poor Soil Quality	1.6602*** (0.2201)	1.5857*** (0.2059)
Solar Irradiance	12.4077*** (1.1024)	57.1575*** (3.8707)
Proximity to Substation (log inv. km)	0.8258*** (0.1433)	0.7776*** (0.1125)
Prox. to Transmission Lines (log inv. km)	0.4680*** (0.0945)	0.5134*** (0.0822)
Parcel Area (log ha)	0.4630*** (0.0337)	0.5398*** (0.0392)
Year Fixed Effects	Yes	No
Observations	3,016,944	3,016,944

Notes: Table reports main effects in non-standardized units: drought index (0–1 scale), farm labor costs (log real weekly wages in USD), soil quality (0–1 scale), solar irradiance (0–1 normalized), proximity to transmission lines and substation (log inverse km), and parcel area (log hectares). Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Column (1) reports corrected LPM coefficients (outcome: solar adoptions per 10,000 parcel-years). Column (2) reports Cox log-hazard coefficients, which are not exponentiated. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Drought Interactions Across Estimators

	LPM (per 10k)	Cox (HR)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	1.1102*** (0.1915)	3.0423*** (1.1417)	0.1108 (0.1524)
Labor Cost (t-1 to t-3, log)	0.5369*** (0.0617)	7.8723*** (1.4094)	0.6063*** (0.0675)
Poor Soil Quality	1.6324*** (0.2173)	4.6895*** (1.0019)	1.4691*** (0.2037)
Solar Irradiance	0.6797*** (0.0603)	24.2429*** (5.4384)	2.1860*** (0.1413)
Proximity to Substation	0.5257*** (0.0918)	1.6481*** (0.1245)	0.4856*** (0.0791)
Prox. to Transmission Lines	0.3245*** (0.0655)	1.4320*** (0.0821)	0.3557*** (0.0575)
Parcel Area (log)	0.9582*** (0.0699)	3.1597*** (0.2855)	1.2553*** (0.0938)
Drought Interactions			
Drought $\times$ Poor Soil	0.6036** (0.2412)	1.2130 (0.2060)	0.2055 (0.1963)
Drought $\times$ Solar Irradiance	0.0804 (0.0538)	0.7275** (0.1029)	-0.0017 (0.1200)
Drought $\times$ Proximity Substation	0.1874** (0.0867)	1.0125 (0.0529)	0.0218 (0.0624)
Drought $\times$ Parcel Area	0.1581** (0.0769)	0.9170 (0.0620)	-0.1327 (0.0812)
Year Fixed Effects	Yes	No	No
Observations	3,016,944	3,016,944	3,016,944

Notes: All models restrict the estimation sample to parcel-years up to the year of solar adoption. Column (1) reports coefficient estimates with standard errors in parentheses for a linear probability model with year fixed effects. The dependent variable is rescaled to solar adoptions per 10,000 parcel-years. Column (2) reports hazard ratios from a Cox proportional hazards model. Column (3) reports coefficient estimates from a complementary log-log model. Logit is omitted from this table since average marginal effects cannot be computed for interaction terms. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Intensive Margin: Solar Area (Ha) Among Adopting Parcels

	Main Effects	+ Interactions
Drought Exposure (t-1 to t-3)	2.8708 (2.7134)	-1.1513 (2.7689)
Labor Cost (t-1 to t-3, log)	7.5105*** (2.3056)	7.0895*** (2.3302)
Poor Soil Quality	1.1805 (2.8516)	1.4262 (2.7323)
Solar Irradiance	8.4300*** (2.4502)	8.1908*** (2.4322)
Proximity to Substation	2.6071*** (0.7548)	2.4123*** (0.7225)
Prox. to Transmission Lines	1.2087 (0.8466)	1.2333 (0.8483)
Parcel Area (log)	8.9181*** (1.1710)	8.6759*** (1.1296)
Drought Interactions		
Drought $\times$ Poor Soil		-0.6623 (1.4031)
Drought $\times$ Solar Irradiance		2.3027** (0.9937)
Drought $\times$ Proximity Substation		1.1605*** (0.3322)
Drought $\times$ Parcel Area		2.0243*** (0.5895)
Year Fixed Effects	Yes	Yes
Sample	Solar Parcels Only	Solar Parcels Only
Observations	4,509	4,509
Adjusted R-squared	0.26811	0.27516

Notes: Table reports OLS estimates with robust standard errors clustered at the parcel level in parentheses. The outcome is solar area in hectares. Sample is restricted to parcels that ever adopt solar, including all post-adoption years, during which solar area may increase as construction continues. All continuous covariates are standardized to mean zero, unit SD. Year fixed effects are included.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Prior Land Use Across Estimators

	Corrected LPM/Logit		Survival Models	
	LPM	Logit (AME)	Cox (HR)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	1.3722*** (0.2175)	0.0000 (0.0000)	1.3679 (0.4418)	0.0121 (0.0678)
Labor Cost (t-1 to t-3, log)	0.7512*** (0.0882)	0.0002*** (0.0000)	6.2099*** (1.1559)	0.5979*** (0.0672)
Poor Soil Quality	1.0774*** (0.2468)	0.0001*** (0.0000)	3.8783*** (0.9114)	1.1987*** (0.2219)
Solar Irradiance	0.6844*** (0.0618)	0.0003*** (0.0000)	21.5586*** (5.7785)	2.1773*** (0.1677)
Proximity to Substation	0.5898*** (0.1022)	0.0001*** (0.0000)	1.6641*** (0.1170)	0.4997*** (0.0713)
Prox. to Transmission Lines	0.3893*** (0.0704)	0.0000*** (0.0000)	1.4600*** (0.0846)	0.3757*** (0.0581)
Parcel Area (log)	1.0856*** (0.0861)	0.0001*** (0.0000)	2.8751*** (0.2528)	1.1365*** (0.0869)
Prior Land Use (t-1)				
Cereal Grains	2.0175*** (0.3708)	0.0002*** (0.0000)	5.1565*** (1.1695)	1.5425*** (0.2225)
Feed, Fiber, Industrial Crops	-0.6479*** (0.1923)	-0.0000 (0.0000)	0.9146 (0.3145)	-0.1584 (0.3394)
Fruits	-0.9633*** (0.1498)	-0.0000*** (0.0000)	0.2339** (0.1435)	-1.7112*** (0.6080)
Vegetables and Melons	0.1088 (0.4710)	0.0000 (0.0000)	1.5791 (0.5985)	0.4708 (0.3770)
Nuts	-0.8149*** (0.1559)	-0.0000* (0.0000)	0.5356* (0.1788)	-0.8450*** (0.3251)
Tree Crops	1.0113 (1.1002)	0.0004 (0.0004)	7.5648** (7.6165)	1.1701 (1.0208)
Fallow/Idle Cropland	2.8542*** (0.3834)	0.0002*** (0.0000)	4.5946*** (0.9120)	1.5238*** (0.1953)
Developed Land	0.3121** (0.1314)	-0.0000 (0.0000)	0.7957 (0.4008)	-0.3125 (0.5001)
Barren/Shrubland	0.9094*** (0.2821)	-0.0000 (0.0000)	0.7094 (0.4407)	-0.2006 (0.6135)
Forest/Wetlands	-0.1653* (0.0909)	0.0000 (.)	0.0000*** (0.0000)	0.0000 (.)
Year Fixed Effects	Yes	Yes	No	No
Observations	2,847,072	2,651,268	2,847,072	2,829,900

Notes: All models restrict the estimation sample to parcel-years up to the year of solar adoption. Reference land use category is Grassland/Pasture. Water parcels are excluded. Columns (1)–(2) include year fixed effects and logit reports average marginal effects. Column (3) reports hazard ratios from Cox proportional hazard model. Column (4) reports cloglog coefficients. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Effects of Groundwater Well Depth

	Corrected LPM (per 10k)	Cox (HR)
Drought Exposure (t-1 to t-3)	1.1083*** (0.2079)	1.0410 (0.3592)
Labor Cost (t-1 to t-3, log)	0.6928*** (0.0693)	7.2262*** (1.3564)
Poor Soil Quality	1.4979*** (0.2146)	8.9429*** (2.3071)
Solar Irradiance	0.4909*** (0.0463)	17.0472*** (4.0571)
Proximity to Substation	0.4553*** (0.0924)	1.6914*** (0.1218)
Prox. to Transmission Lines	0.3763*** (0.0678)	1.4128*** (0.0830)
Parcel Area (log)	0.9095*** (0.0677)	2.8847*** (0.2399)
Well Depth (IDW, std)	1.2171*** (0.1314)	1.5120*** (0.0459)
Year Fixed Effects	Yes	No
Observations	3,016,944	3,016,944

Notes: Replicates Table 1 adding an inverse-distance-weighted measure of groundwater well depth. Column (1) reports coefficient estimates with standard errors in parentheses from a corrected LPM with year fixed effects. Outcome is rescaled to solar adoptions per 10,000 parcel-years. Column (2) reports hazard ratios from a Cox proportional hazards model. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Robustness: Logit and Cloglog, Groundwater Well Depth

	Logit (AME)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	0.0000 (0.0000)	0.0635 (0.0663)
Labor Cost (t-1 to t-3, log)	0.0002*** (0.0000)	0.5954*** (0.0663)
Poor Soil Quality	0.0002*** (0.0000)	2.0973*** (0.2457)
Solar Irradiance	0.0003*** (0.0000)	1.7316*** (0.1464)
Proximity to Substation	0.0001*** (0.0000)	0.5139*** (0.0740)
Prox. to Transmission Lines	0.0000*** (0.0000)	0.3500*** (0.0590)
Parcel Area (log)	0.0001*** (0.0000)	1.1480*** (0.0830)
Well Depth (IDW, std)	0.0000*** (0.0000)	0.4405*** (0.0325)
Year Fixed Effects	Yes	No
Observations	2,667,739	3,016,944

Notes: Replicates Appendix Table A13 adding groundwater well depth. Column (1) reports average marginal effects from a logit model with year fixed effects. Column (2) reports coefficients from a cloglog model without year fixed effects. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Prime, Important, and Unique Farmland

	Corrected LPM (per 10k)	Cox (HR)
Drought Exposure (t-1 to t-3)	1.3364*** (0.2106)	1.9382** (0.5760)
Labor Cost (t-1 to t-3, log)	0.5064*** (0.0568)	5.7277*** (1.0122)
Poor Soil Quality	0.3968* (0.2258)	1.7242** (0.4200)
Solar Irradiance	0.6354*** (0.0568)	14.2876*** (3.2129)
Proximity to Substation	0.5147*** (0.0916)	1.6215*** (0.1179)
Prox. to Transmission Lines	0.3975*** (0.0678)	1.4839*** (0.0860)
Parcel Area (log)	1.4028*** (0.1032)	3.5053*** (0.2956)
Prime Farmland	-1.6864*** (0.1609)	0.2925*** (0.0483)
Important Farmland	-0.1244 (0.1297)	1.5035*** (0.1951)
Unique Farmland	-1.0720*** (0.1412)	0.4359*** (0.1005)
Year Fixed Effects	Yes	No
Observations	3,016,944	3,016,944

Notes: Replicates Table 1 adding indicators for Prime, Unique, and Important farmland, based on USDA farmland classification data. Column (1) is a corrected LPM with year fixed effects. Outcome is rescaled to solar adoptions per 10,000 parcel-years. Column (2) reports hazard ratios from a Cox proportional hazards model. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Robustness: Logit and Cloglog, Prime, Important, and Unique Farmland

	Logit (AME)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	0.0001** (0.0000)	0.0766 (0.0654)
Labor Cost (t-1 to t-3, log)	0.0002*** (0.0000)	0.5668*** (0.0657)
Poor Soil Quality	0.0001** (0.0000)	0.4026* (0.2338)
Solar Irradiance	0.0003*** (0.0000)	1.8477*** (0.1361)
Proximity to Substation	0.0000*** (0.0000)	0.4663*** (0.0738)
Prox. to Transmission Lines	0.0000*** (0.0000)	0.4012*** (0.0580)
Parcel Area (log)	0.0001*** (0.0000)	1.3567*** (0.0807)
Prime Farmland	-0.0001*** (0.0000)	-1.3578*** (0.1639)
Important Farmland	0.0000*** (0.0000)	0.3582*** (0.1279)
Unique Farmland	-0.0001*** (0.0000)	-0.9240*** (0.2302)
Year Fixed Effects	Yes	No
Observations	2,667,739	3,016,944

Notes: Replicates Appendix Table A13 adding indicators for Prime, Important, and Unique farmland. Column (1) reports average marginal effects from a logit model with year fixed effects. Column (2) reports coefficients from a cloglog model without year fixed effects. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Spatial Spillovers: Neighboring Solar Development

	Indicator: Any Solar Within Radius			Count: Solar Facilities Within Radius		
	2km	5km	10km	2km	5km	10km
Drought Exposure (t-1 to t-3)	1.3978*** (0.2124)	1.4590*** (0.2150)	1.4530*** (0.2152)	1.4230*** (0.2128)	1.4062*** (0.2120)	1.3826*** (0.2112)
Labor Cost (t-1 to t-3, log)	0.5578*** (0.0587)	0.5544*** (0.0582)	0.5642*** (0.0602)	0.5718*** (0.0597)	0.5730*** (0.0598)	0.5698*** (0.0595)
Poor Soil Quality	1.6608*** (0.2201)	1.6532*** (0.2197)	1.6564*** (0.2194)	1.6698*** (0.2208)	1.6783*** (0.2216)	1.6817*** (0.2214)
Solar Irradiance	0.6649*** (0.0592)	0.6882*** (0.0612)	0.6830*** (0.0606)	0.6625*** (0.0590)	0.6497*** (0.0584)	0.6405*** (0.0580)
Proximity to Substation	0.5206*** (0.0921)	0.5301*** (0.0920)	0.5299*** (0.0921)	0.5294*** (0.0920)	0.5346*** (0.0921)	0.5405*** (0.0921)
Prox. to Transmission Lines	0.3249*** (0.0656)	0.3224*** (0.0654)	0.3244*** (0.0656)	0.3264*** (0.0657)	0.3207*** (0.0653)	0.3182*** (0.0652)
Parcel Area (log)	0.9585*** (0.0699)	0.9625*** (0.0701)	0.9647*** (0.0702)	0.9627*** (0.0701)	0.9628*** (0.0701)	0.9651*** (0.0702)
Neighbor Solar (Indicator, t-1)						
Any Solar Neighbor, 2km (t-1)	0.6053*** (0.1448)					
Any Solar Neighbor, 5km (t-1)		0.3021** (0.1391)				
Any Solar Neighbor, 10km (t-1)			0.0970 (0.1675)			
Neighbor Solar (Count, t-1)						
Solar Neighbor Count, 2km (t-1)				0.0382*** (0.0088)		
Solar Neighbor Count, 5km (t-1)					0.0225*** (0.0062)	
Solar Neighbor Count, 10km (t-1)						0.0174*** (0.0041)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,016,944	3,016,944	3,016,944	3,016,944	3,016,944	3,016,944
Adjusted R-squared	0.00031	0.00030	0.00030	0.00030	0.00030	0.00031

Notes: Each column adds a measure of neighboring solar development to the main effects LPM specification. Neighbor solar measures are lagged one year ($t - 1$) to capture lagged responses. Columns (1)–(3) use an indicator for whether any utility-scale solar facility existed within the given radius as of $t - 1$. Columns (4)–(6) use the count of such facilities. All models restrict the estimation sample to parcel-years up to the year of solar adoption and include year fixed effects. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A13: Robustness: Logit and Cloglog (Main Effects)

	Logit (AME)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	0.0001** (0.0000)	0.0705 (0.0660)
Labor Cost (t-1 to t-3, log)	0.0002*** (0.0000)	0.6050*** (0.0676)
Poor Soil Quality	0.0002*** (0.0000)	1.5278*** (0.1970)
Solar Irradiance	0.0003*** (0.0000)	2.1885*** (0.1360)
Proximity to Substation	0.0000*** (0.0000)	0.4910*** (0.0739)
Prox. to Transmission Lines	0.0000*** (0.0000)	0.3554*** (0.0575)
Parcel Area (log)	0.0001*** (0.0000)	1.2141*** (0.0828)
Year Fixed Effects	Yes	No
Observations	2,667,739	3,016,944

Notes: Column (1) reports average marginal effects from a logit model with year fixed effects. Column (2) reports coefficients from a complementary log-log (cloglog) model without year fixed effects. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A14: Robustness: County and Parcel Fixed Effects

	Corrected LPM (per 10k)			Cox
	County FE	Parcel FE	Parcel FE + Int.	County FE (HR)
Drought Exposure (t-1 to t-3)	-0.2597 (0.2194)	-0.3030 (0.2091)	-0.0388 (0.2190)	0.5108 (0.2626)
Labor Cost (t-1 to t-3, log)	0.9330*** (0.1026)	1.2551*** (0.1123)	0.1836 (0.1232)	17.9216*** (8.7358)
Poor Soil Quality	2.0835*** (0.2588)			4.8052*** (1.0945)
Solar Irradiance	0.1971*** (0.0272)			10.6321*** (3.7450)
Proximity to Substation	0.6687*** (0.1032)			1.6567*** (0.1194)
Prox. to Transmission Lines	0.3590*** (0.0691)	0.0000 (.)	0.0000 (.)	1.4942*** (0.0888)
Parcel Area (log)	1.0576*** (0.0757)			3.2875*** (0.2743)
Drought Interactions				
Drought × Poor Soil			0.0997 (0.2574)	
Drought × Solar Irradiance			-0.0243 (0.0655)	
Drought × Proximity Substation			0.3055*** (0.0950)	
Drought × Parcel Area			0.0399 (0.0915)	
Labor Interactions				
Labor × Poor Soil			2.4147*** (0.3261)	
Labor × Solar Irradiance			1.0552*** (0.1078)	
Labor × Proximity Substation			0.4919*** (0.0989)	
Labor × Parcel Area			1.2146*** (0.1163)	
Year Fixed Effects	Yes	Yes	Yes	No
County Fixed Effects	Yes	No	No	Yes
Parcel Fixed Effects	No	Yes	Yes	No
Observations	3,016,944	3,455,182	3,016,944	3,016,944

Notes: All models restrict the estimation sample to parcel-years up to the year of solar adoption. Column (1) adds county fixed effects to the corrected LPM. Column (2) adds parcel fixed effects, such that time-invariant covariates (soil quality, irradiance, infrastructure, area) are absorbed and omitted. Column (3) adds interactions between time-varying drought and labor costs and time-invariant parcel characteristics. Column (4) reports hazard ratios from a Cox proportional hazards model with county fixed effects. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A15: Robustness: Rural-Urban Continuum Code Fixed Effects

	Corrected LPM/Logit		Survival Models	
	LPM	Logit (AME)	Cox (HR)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	1.3600*** (0.2057)	0.0000 (0.0000)	1.4625 (0.5219)	0.0758 (0.0653)
Labor Cost (t-1 to t-3, log)	0.6732*** (0.0866)	0.0002*** (0.0000)	6.6194*** (1.2483)	0.5336*** (0.0606)
Poor Soil Quality	1.5956*** (0.2226)	0.0001*** (0.0000)	4.4140*** (0.9604)	1.3910*** (0.2080)
Solar Irradiance	0.6458*** (0.0597)	0.0003*** (0.0000)	25.7493*** (5.3195)	2.6163*** (0.1686)
Proximity to Substation	0.5749*** (0.0994)	0.0001*** (0.0000)	1.6682*** (0.1174)	0.5024*** (0.0716)
Prox. to Transmission Lines	0.3175*** (0.0666)	0.0000*** (0.0000)	1.4620*** (0.0852)	0.3821*** (0.0585)
Parcel Area (log)	0.9853*** (0.0717)	0.0001*** (0.0000)	3.0835*** (0.2464)	1.1908*** (0.0808)
County Became More Urban, 2003–2023	-0.1757 (0.1717)	0.0000 (0.0000)	1.1976 (0.1688)	0.1689 (0.1574)
RUCC 2003 (ref: Rural, Categories 6–8 Combined)				
Metro, 1M+ Population	-0.3016 (0.1992)		1.2084 (1.2672)	0.9261 (1.0433)
Metro, 250K–1M Population	0.1789* (0.0996)		0.7407 (0.7586)	-0.0783 (1.0229)
Metro, <250K Population	1.0737*** (0.1961)		1.7170 (1.7439)	0.9373 (1.0185)
Nonmetro, Urban Pop. 20K+, Metro-Adjacent	0.0987 (0.1765)		0.3024 (0.4483)	-0.0468 (1.4198)
Year Fixed Effects	Yes	Yes	No	No
RUCC 2003 Fixed Effects	Yes	Yes	Yes	Yes
Observations	3,016,944	2,667,739	3,016,944	3,016,944

Notes: Replicates Table 1 adding Rural-Urban Continuum Code (RUCC) fixed effects based on each county's 2003 classification, as well as a binary indicator for whether the county's RUCC classification became more urban between 2003 and 2023. RUCC categories 6 through 8 (nonmetro, rural) are combined into a single omitted reference category due to limited representation among Central Valley counties. Column (1) reports coefficient estimates for a corrected LPM with year fixed effects. Outcome is rescaled to solar adoptions per 10,000 parcel-years. Column (2) reports average marginal effects from a logit model with year fixed effects. Column (3) reports hazard ratios from a Cox proportional hazards model. Column (4) reports coefficients from a cloglog model. All models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A16: Falsification: All-Sector Average Wage

	Corrected LPM (per 10k)	Cox (HR)
Drought Exposure (t-1 to t-3)	0.9908*** (0.1793)	1.5102 (0.5144)
Total Avg. Wage, log (std)	-0.3554*** (0.0698)	0.6412*** (0.0987)
Poor Soil Quality	1.8466*** (0.2402)	4.9076*** (0.9941)
Solar Irradiance	0.4677*** (0.0406)	6.4431*** (0.8466)
Proximity to Substation	0.5737*** (0.0979)	1.6555*** (0.1235)
Prox. to Transmission Lines	0.3400*** (0.0694)	1.4430*** (0.0849)
Parcel Area (log)	1.0088*** (0.0737)	3.4943*** (0.2967)
Year Fixed Effects	Yes	No
Wage Measure	Total (all-sector)	Total (all-sector)
Observations	2,896,958	2,896,958

Notes: Replicates Table 1 replacing the average local farm wage with the average all-sector wage (measured at the county level). All models restrict the estimation sample to parcel-years up to the year of solar adoption. Continuous covariates are standardized. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A17: Robustness: Measuring Prior Land Use at t-2

	Corrected LPM/Logit		Survival Models	
	LPM (per 10k)	Logit (AME)	Cox (HR)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	1.4332*** (0.2215)	0.0000 (0.0000)	1.2632 (0.4103)	0.0019 (0.0670)
Labor Cost (t-1 to t-3, log)	0.7675*** (0.0987)	0.0002*** (0.0000)	6.8590*** (1.3989)	0.5358*** (0.0713)
Poor Soil Quality	1.3355*** (0.2590)	0.0002*** (0.0000)	4.2259*** (0.9638)	1.2716*** (0.2139)
Solar Irradiance	0.7097*** (0.0647)	0.0004*** (0.0000)	27.3838*** (8.2832)	2.2456*** (0.1785)
Proximity to Substation	0.6100*** (0.1080)	0.0001*** (0.0000)	1.6778*** (0.1208)	0.5122*** (0.0733)
Prox. to Transmission Lines	0.3954*** (0.0755)	0.0000*** (0.0000)	1.4582*** (0.0854)	0.3699*** (0.0588)
Parcel Area (log)	1.1302*** (0.0937)	0.0001*** (0.0000)	2.8820*** (0.2730)	1.1413*** (0.0924)
Prior Land Use (t-2)				
Cereal Grains	1.3951*** (0.3475)	0.0002*** (0.0000)	4.8523*** (1.2605)	1.4451*** (0.2440)
Feed, Fiber, Industrial Crops	0.2853 (0.2829)	0.0001*** (0.0000)	2.6956*** (0.7561)	0.9488*** (0.2705)
Fruits	-0.8215*** (0.1594)	-0.0000 (0.0000)	0.5633 (0.2601)	-0.8480* (0.4510)
Vegetables and Melons	2.1285*** (0.7508)	0.0002*** (0.0000)	4.1105*** (1.2298)	1.4566*** (0.2931)
Nuts	-0.7623*** (0.1441)	-0.0000** (0.0000)	0.3938** (0.1654)	-1.2594*** (0.4071)
Tree Crops	0.3222 (1.1569)	0.0002 (0.0002)	4.4871 (4.6803)	1.2781 (1.0226)
Fallow/Idle Cropland	2.6906*** (0.3920)	0.0002*** (0.0000)	4.3690*** (0.9365)	1.5471*** (0.2072)
Developed Land	0.4619*** (0.1445)	0.0000 (0.0000)	1.1787 (0.5570)	0.0730 (0.4697)
Barren/Shrubland	0.7246*** (0.2042)	-0.0000** (0.0000)	0.2468 (0.2545)	-1.2968 (1.0235)
Forest/Wetlands	-0.1365 (0.0909)	0.0000 (.)	0.0000*** (0.0000)	0.0000 (.)
Year Fixed Effects	Yes	Yes	No	No
Observations	2,667,737	2,473,348	2,667,737	2,651,520

Notes: Table replicates the main land use analysis using prior land use measured at t-2 rather than t-1. Reference category is Grassland/Pasture. Water parcels are excluded. All models restrict the estimation sample to parcel-years up to the year of solar adoption. Column (1) reports coefficient estimates from a corrected LPM with year fixed effects. The outcome is solar adoptions per 10,000 parcel-years. Column (2) reports logit average marginal effects. Column (3) reports hazard ratios from Cox proportional hazards model. Column (4) reports cloglog coefficients. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A18: Robustness: 2014 Substation Distance

	Corrected LPM (per 10k)	Cox (HR)
Drought Exposure (t-1 to t-3)	1.4859*** (0.2145)	1.8854** (0.5816)
Labor Cost (t-1 to t-3, log)	0.5753*** (0.0619)	6.9136*** (1.1292)
Poor Soil Quality	1.6858*** (0.2191)	4.8918*** (1.0028)
Solar Irradiance	0.7009*** (0.0611)	23.6115*** (4.8629)
Proximity to Substation, 2014	0.6281*** (0.1366)	1.3092*** (0.0655)
Prox. to Transmission Lines	0.3327*** (0.0764)	1.5997*** (0.0793)
Parcel Area (log)	0.9392*** (0.0686)	2.8078*** (0.2152)
Year Fixed Effects	Yes	No
Observations	3,018,891	3,018,891

Notes: Replicates Table 1 replacing current-period substation proximity measure with distance to the nearest substation as of 2014, prior to the bulk of utility-scale solar development in the sample. Column (1) reports coefficient estimates with standard errors in parentheses for a corrected LPM with year fixed effects. Outcome is rescaled to solar adoptions per 10,000 parcel-years. Column (2) reports hazard ratios from a Cox proportional hazards model. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A19: Robustness: Logit and Cloglog, 2014 Substation Distance

	Logit (AME)	Cloglog (Coef.)
Drought Exposure (t-1 to t-3)	0.0001** (0.0000)	0.0785 (0.0656)
Labor Cost (t-1 to t-3, log)	0.0002*** (0.0000)	0.5928*** (0.0671)
Poor Soil Quality	0.0002*** (0.0000)	1.5083*** (0.1958)
Solar Irradiance	0.0003*** (0.0000)	2.2452*** (0.1337)
Proximity to Substation, 2014	0.0000*** (0.0000)	0.2738*** (0.0508)
Prox. to Transmission Lines	0.0000*** (0.0000)	0.4618*** (0.0500)
Parcel Area (log)	0.0001*** (0.0000)	1.1137*** (0.0780)
Year Fixed Effects	Yes	No
Observations	2,669,511	3,018,891

Notes: Replicates Appendix Table A13 replacing the current-period substation proximity measure with distance to the nearest substation as of 2014. Column (1) reports average marginal effects from a logit model with year fixed effects. Column (2) reports coefficients from a cloglog model without year fixed effects. Both models restrict the estimation sample to parcel-years up to the year of solar adoption. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Supplementary Figures

Figure A1: Parcel Distance from Electrical Transmission Lines and Substations

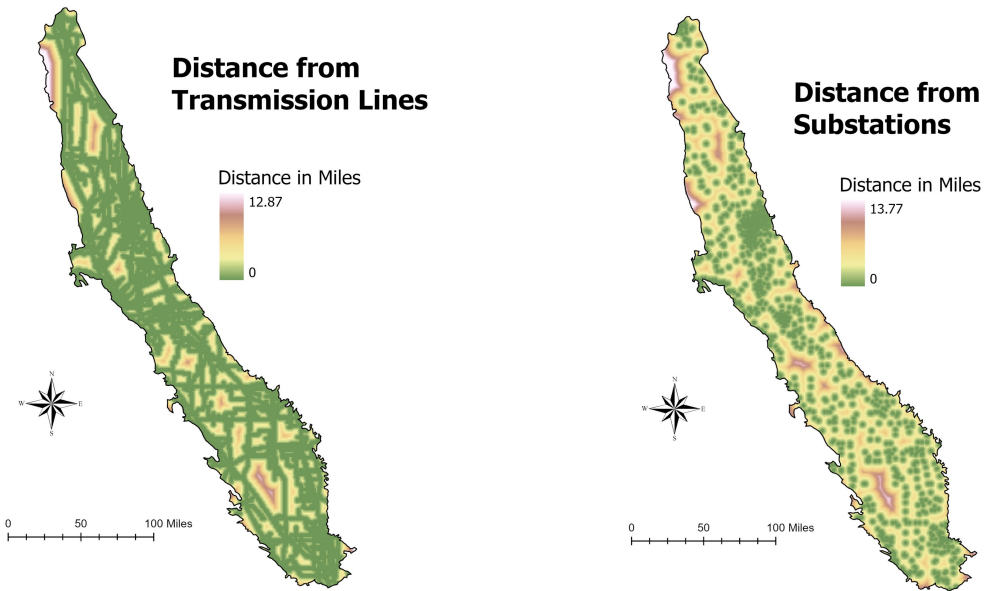
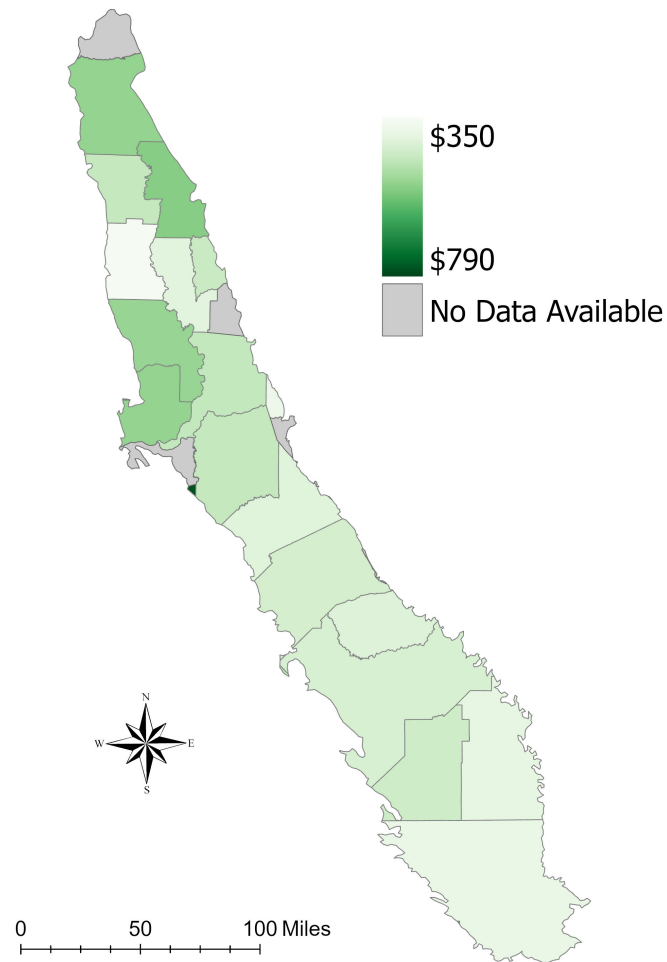
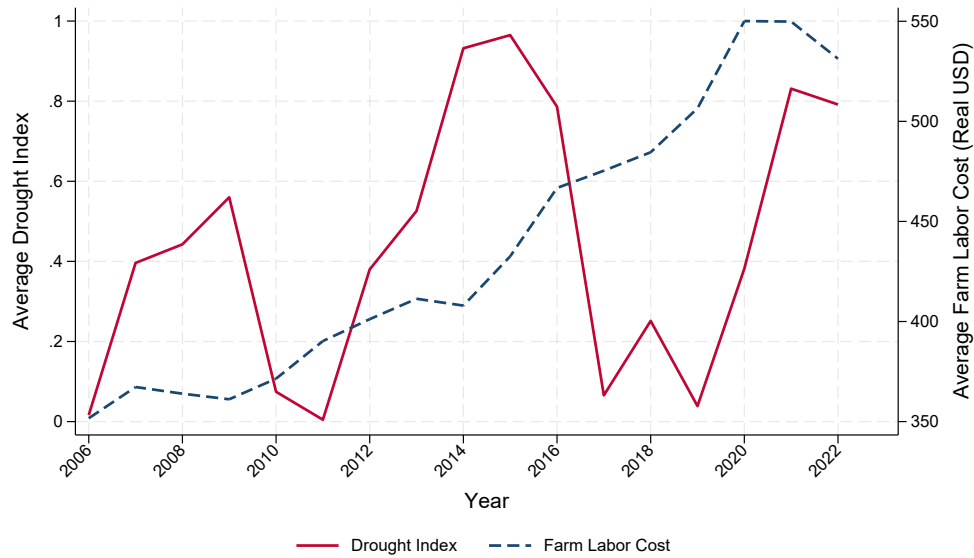


Figure A2: Average Farm Labor Costs (2006-2023)



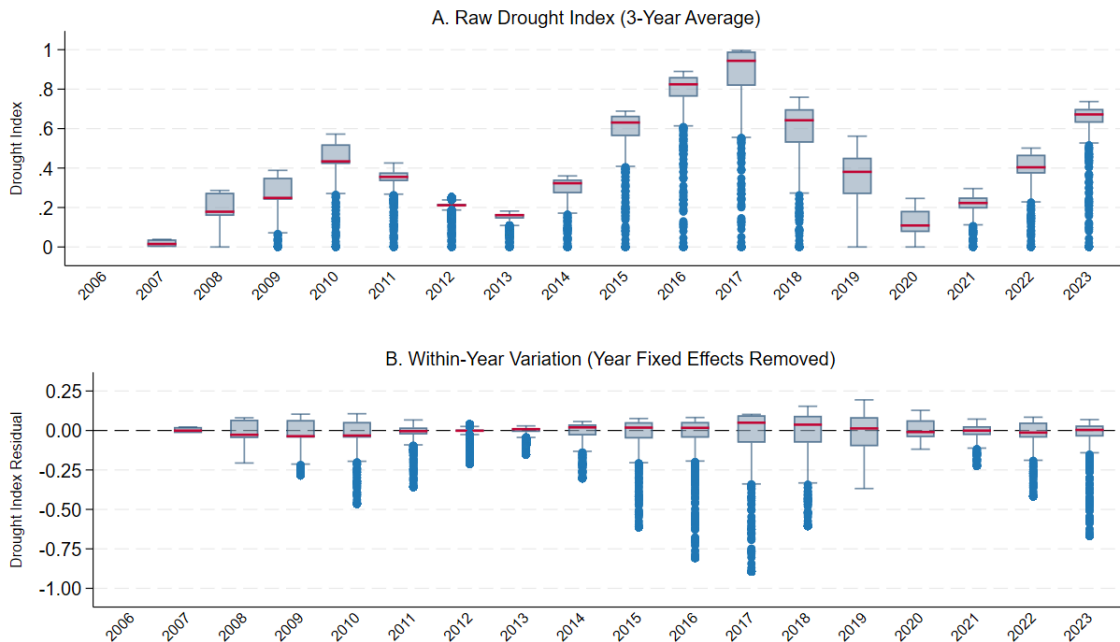
Notes: Figure maps county-level real weekly wage for farm labor contractors and crew leaders (NAICS code 115115), averaged across the 2006-2023 period. Data are not available for some county-years.

Figure A3: Average Drought Exposure and Farm Labor Costs



Notes: Figure plots average drought index (left axis) and average farm labor cost in real USD (right axis) for the Central Valley, by year. Both series are averaged across all parcels in the sample for each year.

Figure A4: Within-Year Variation in Drought Exposure Across Parcels



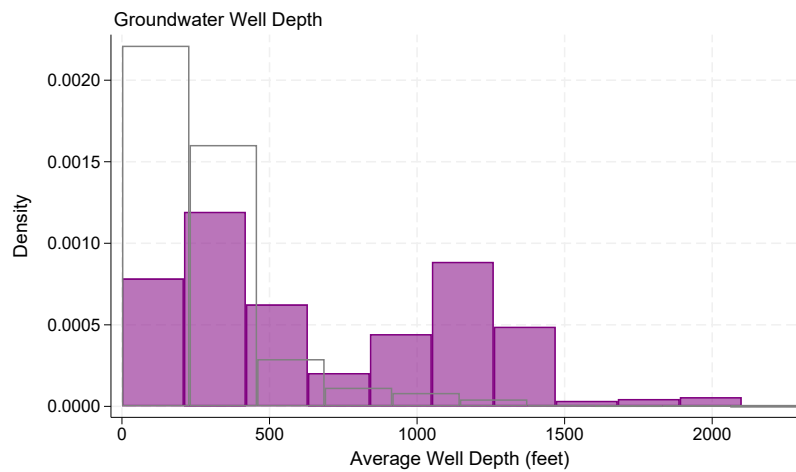
Notes: Panel A shows the distribution of parcel-level drought exposure (3-year average drought index, 0 = no drought, 1 = severe drought) within each year. Panel B removes the common year-level time trend via year fixed effects; remaining variation is purely cross-sectional across parcels within each year.

Figure A5: Within-Year Variation in Labor Costs Across Parcels



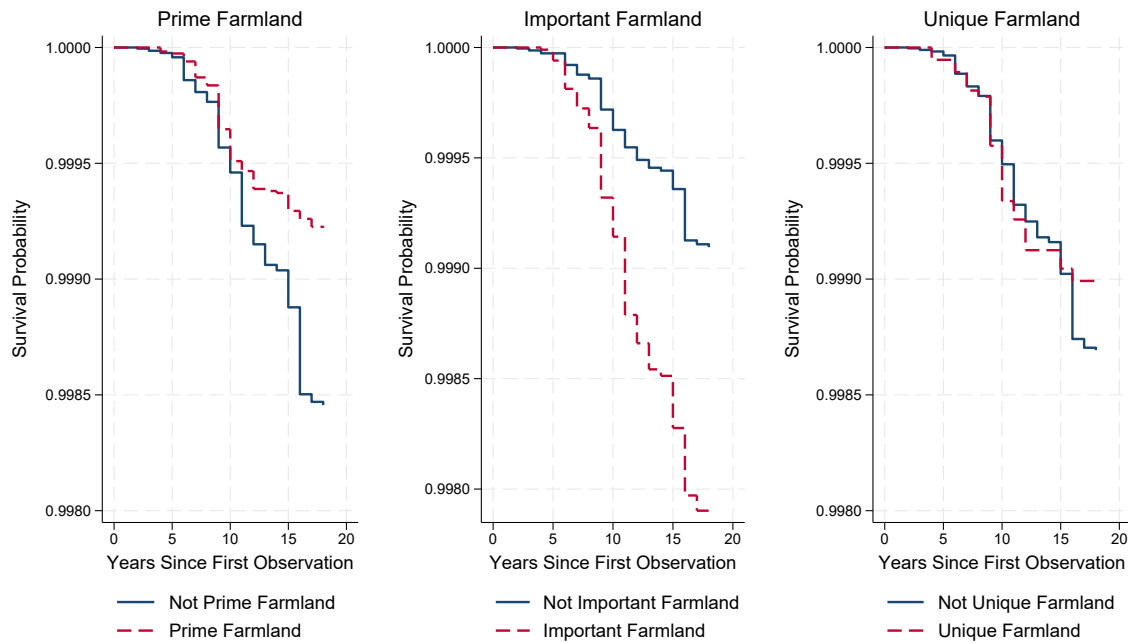
Notes: Panel A shows the distribution of parcel-level labor costs (log, 3-year average) within each year. Panel B removes the common year-level time trend via year fixed effects; remaining variation is purely cross-sectional across counties within each year.

Figure A6: Distribution of Average Well Depth: Solar vs. Non-Solar Parcels



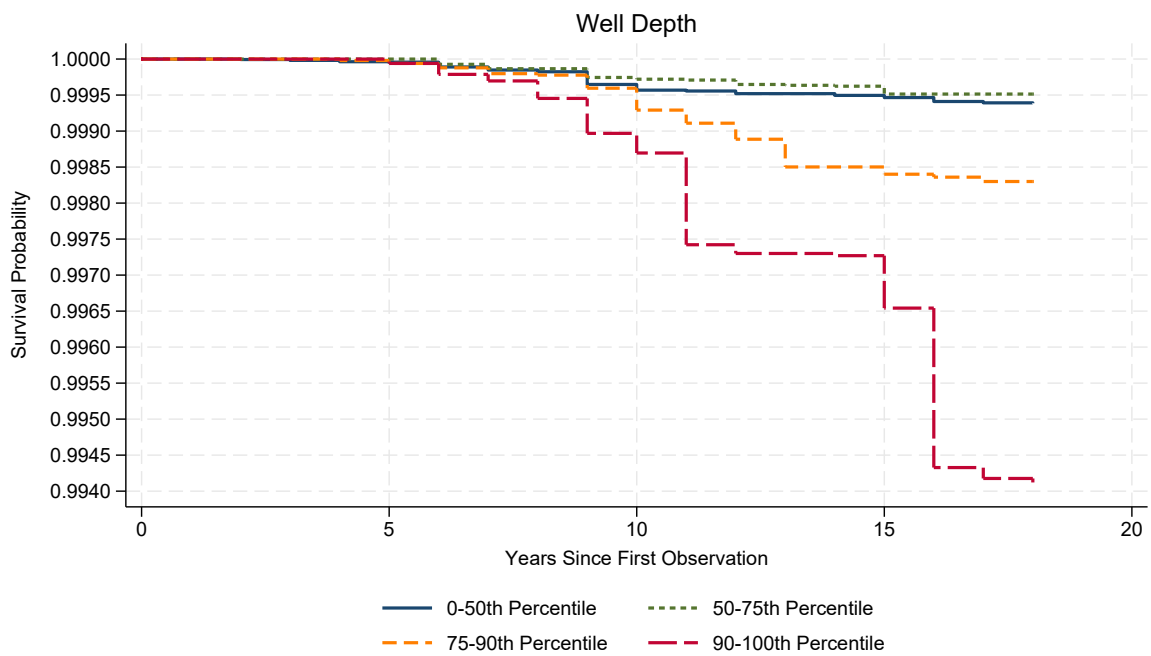
Notes: Histogram of average nearby well depth (feet) for parcels hosting utility-scale solar facilities (colored bars) compared to non-solar parcels (gray outlines).

Figure A7: Kaplan-Meier Survival Curves: Prime, Important, and Unique Farmland



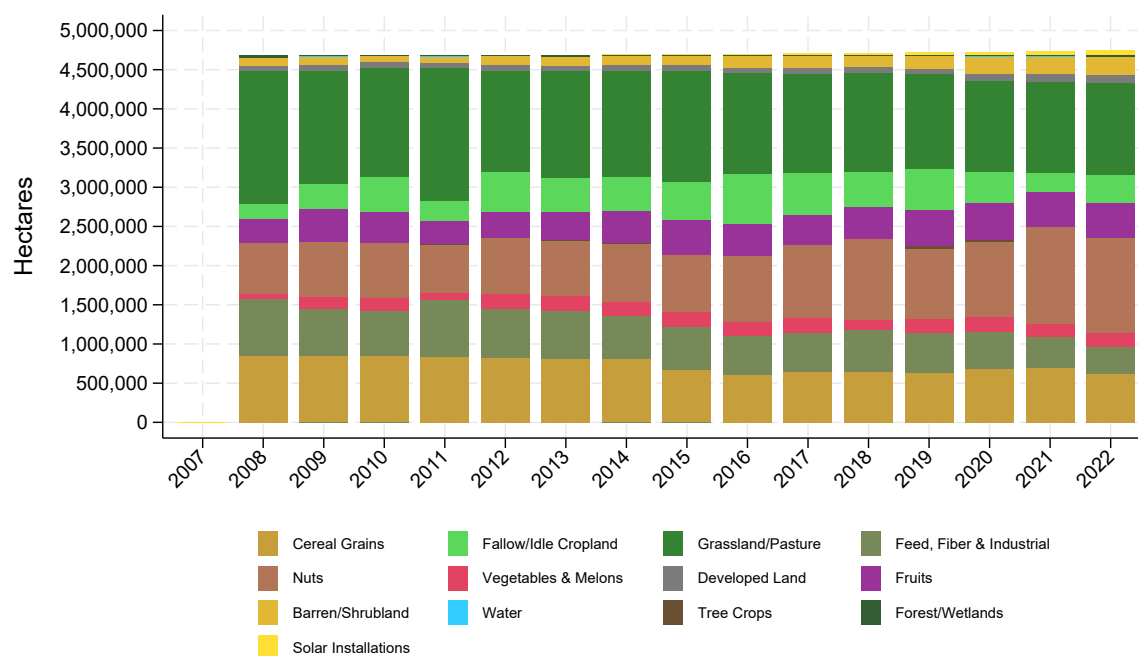
Notes: Kaplan-Meier survival curves comparing time to solar adoption for parcels classified as Prime farmland, Important farmland (statewide, local, or local-potential importance), and Unique farmland (specialty crops enabled by local microclimate), versus parcels not classified under each category, based on USDA farmland classification data. Survival probability reflects the share of parcels that have not yet adopted solar as of each year since first observation.

Figure A8: Kaplan-Meier Survival Curves: Groundwater Well Depth



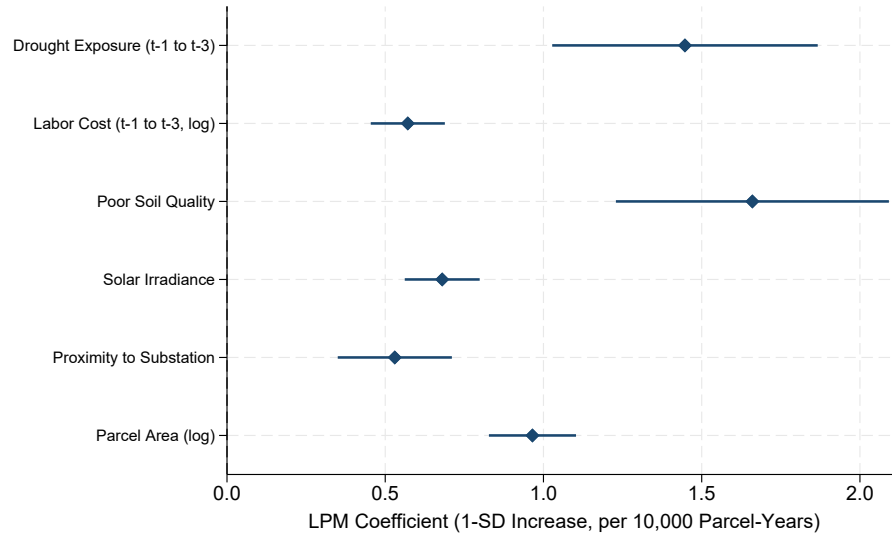
Notes: Figure compares time to solar adoption across four groups of parcels based on well depth: the bottom half of the sample (0–50th percentile), the third quartile (50–75th percentile), the 75–90th percentile, and the top decile (90–100th percentile). Well depth is an inverse-distance-weighted average of nearby well readings.

Figure A9: Total Agricultural Land Use and Cumulative Solar Area, 2007–2022



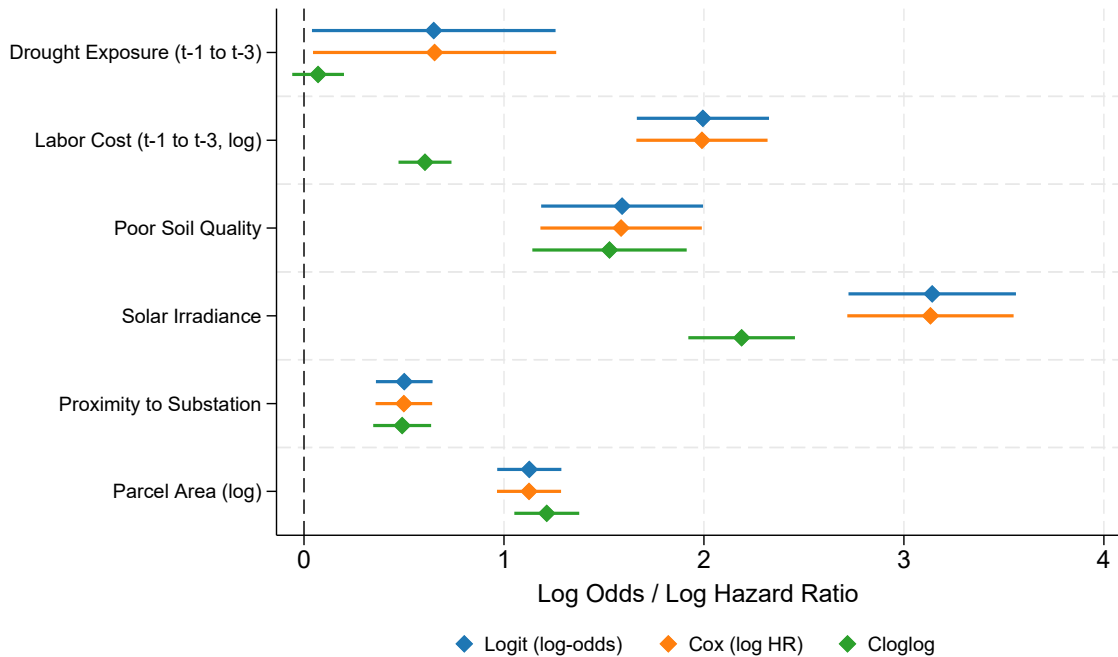
Notes: Figure shows total hectares by land use category across all Central Valley parcels in each year, with cumulative solar area (yellow) shown on top. Land use categories are based on the USDA Cropland Data Layer (CDL). Solar area represents the cumulative stock of utility-scale solar installed as of each year across all parcels in the sample. The near-invisible solar slice relative to the agricultural land base illustrates that utility-scale solar development occupies a minuscule fraction of total Central Valley agricultural land over the study period.

Figure A10: Standardized Coefficients



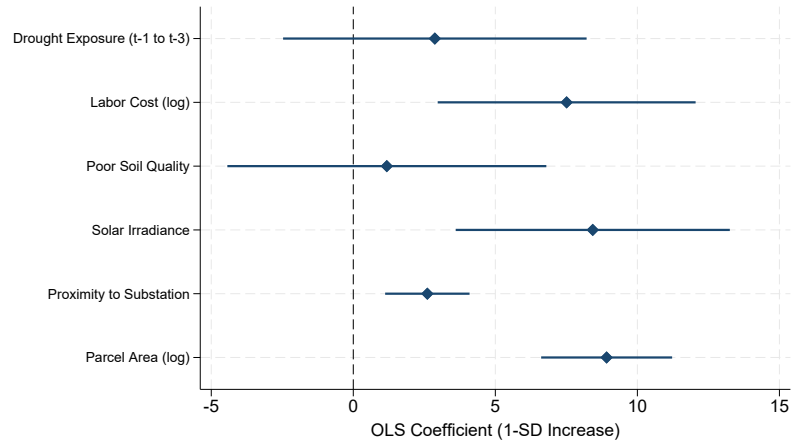
Notes: Figure plots standardized coefficients from Column 1 of Table 1. Each coefficient represents the change in solar adoption probability (in standard deviations) resulting from a one standard deviation increase in the predictor variable. Error bars show 95% confidence intervals based on robust standard errors clustered at the parcel level.

Figure A11: Determinants of Solar Adoption Across Estimators



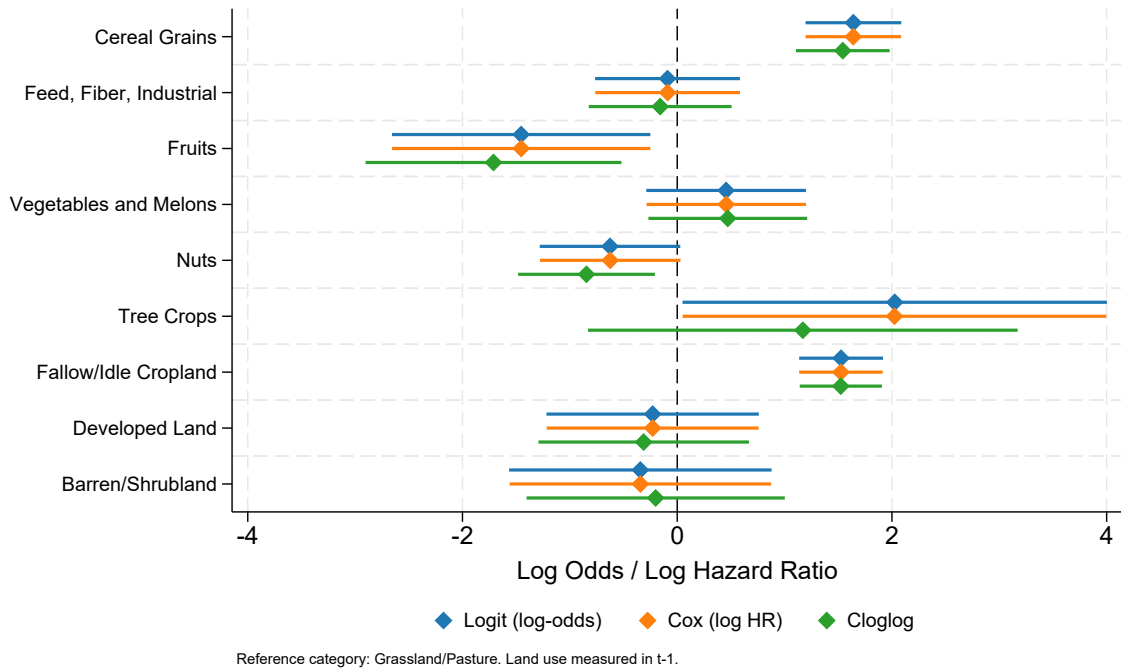
Notes: Figure plots coefficients from three estimators on the corrected risk set (post-adoption observations excluded): log-odds coefficients from a logit model with year fixed effects, log hazard ratios from a Cox proportional hazards model, and coefficients from a complementary log-log (cloglog) model, both without year fixed effects. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

Figure A12: Intensive Margin: Solar Area (Ha) on Adopting Parcels



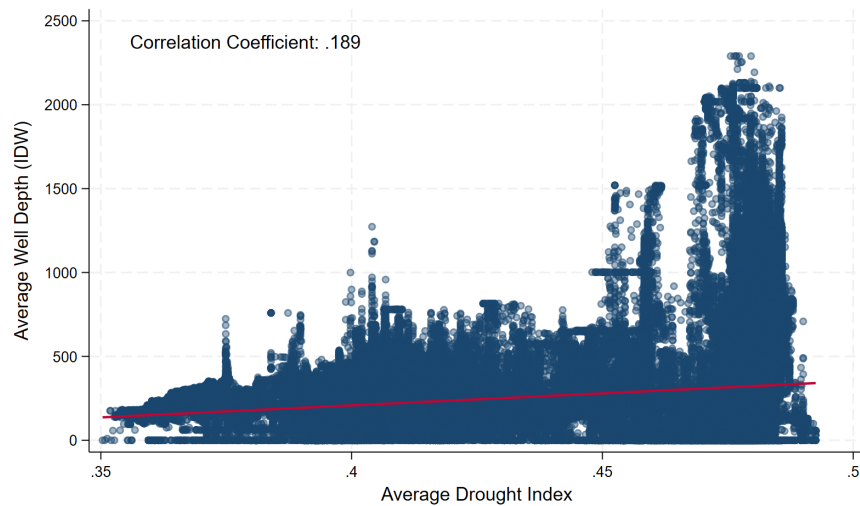
Notes: Coefficient plot from OLS regression of installed solar area (hectares) on parcel characteristics, restricted to solar-adopting parcels. Points represent coefficient estimates with 95% confidence intervals. Robust standard errors clustered at the parcel level.

Figure A13: Prior Land Use and Solar Adoption Across Estimators



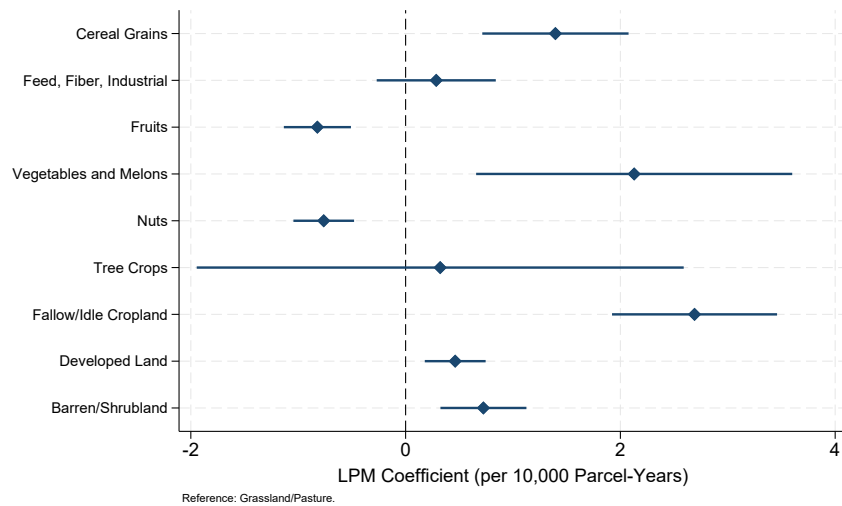
Notes: Figure plots coefficients for prior land use categories (CDL t-1) from three estimators on the corrected risk set: log-odds coefficients from a logit model with year fixed effects, log hazard ratios from Cox, and coefficients from cloglog. Reference category is Grassland/Pasture. Forest/Wetlands excluded from plot due to near-zero adoption rate producing unstable Cox estimates. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

Figure A14: Correlation Between Well Depth and Drought Exposure



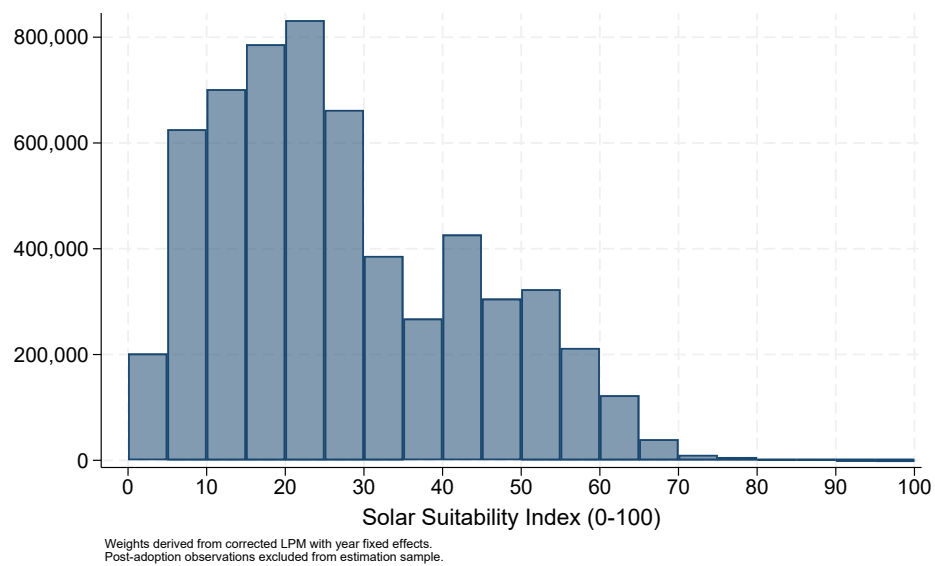
Notes: Figure plots parcel-level averages of well depth (inverse-distance-weighted) against parcel-level averages of the drought index over the full sample period, 2006–2023. Each point represents one parcel. The line shows the linear fit.

Figure A15: Prior Land Use at t-2 and Solar Adoption (Corrected LPM)



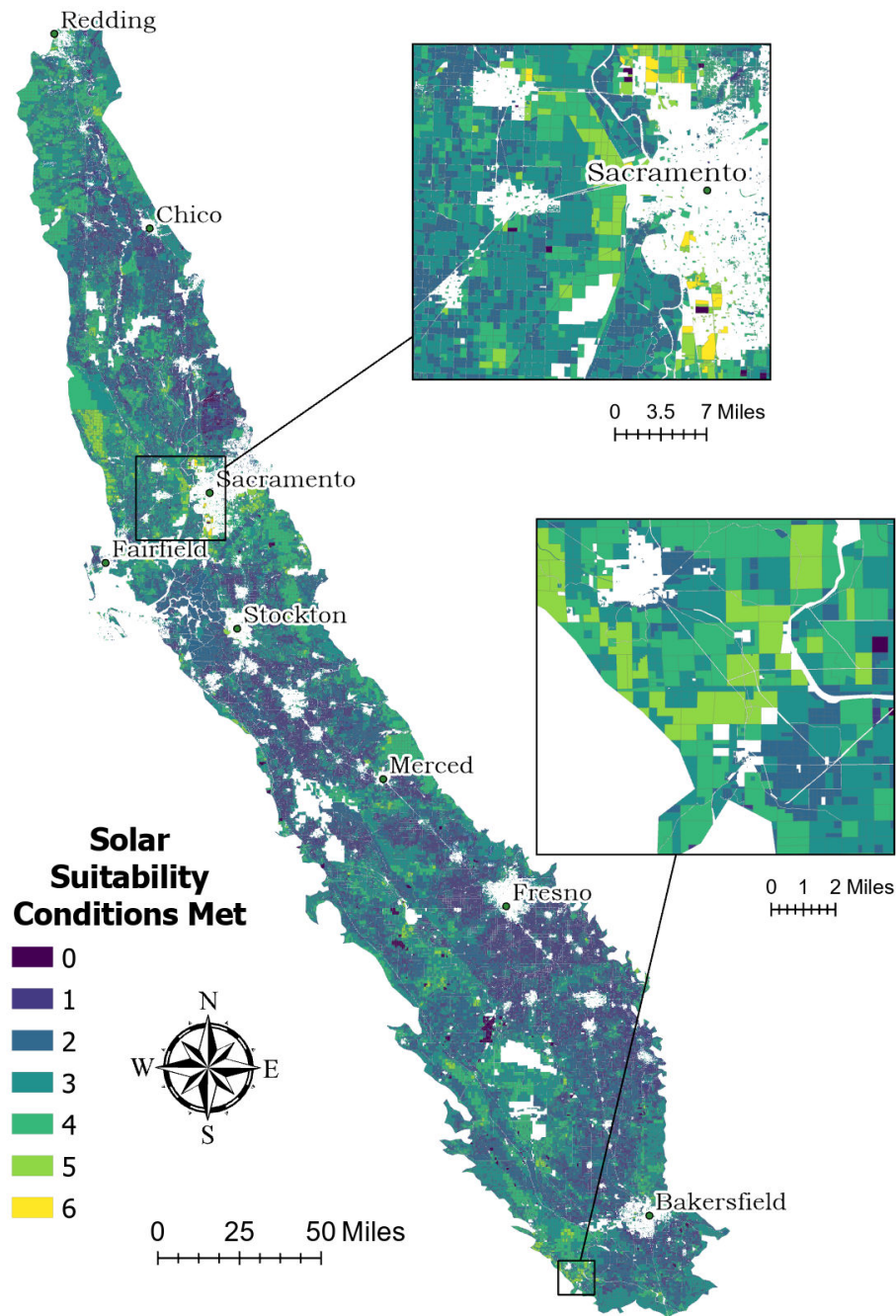
Notes: Figure replicates the main land use coefficient plot using prior land use measured at t-2 rather than t-1. Coefficients are from a corrected LPM with year fixed effects. The dependent variable is solar adoptions per 10,000 parcel-years. Reference category is Grassland/Pasture. Forest/Wetlands omitted for scale. Post-adoption observations are excluded. All continuous covariates are standardized to mean zero, unit SD. Standard errors are clustered at the parcel level.

Figure A16: Distribution of Solar Suitability Scores



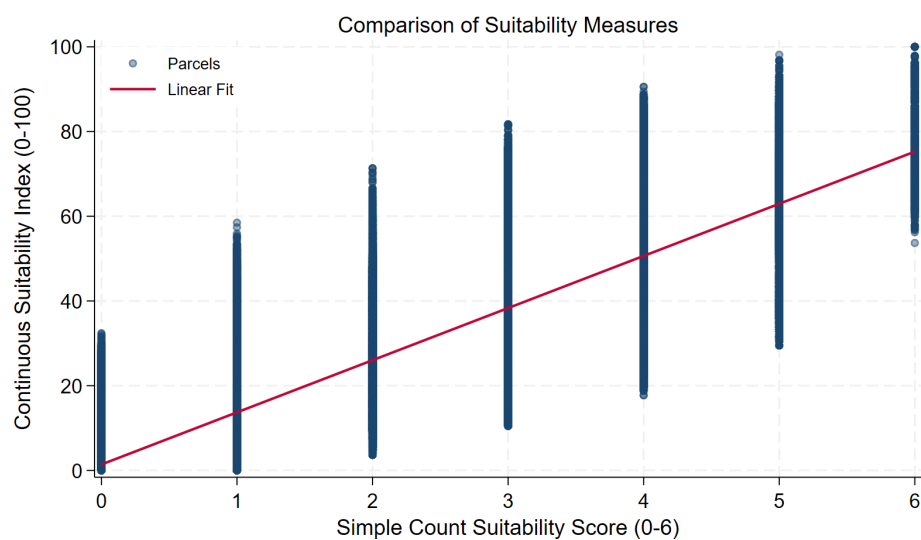
Notes: Histogram shows the distribution of solar suitability scores for non-solar parcels in California's Central Valley. The suitability index (0-100) is a weighted composite of five factors: proximity to electrical substations, parcel size, drought exposure, solar irradiance, and poor soil quality. Higher scores indicate parcels that are both technically suitable for solar development and economically marginal for agriculture.

Figure A17: Simple Solar Suitability Index



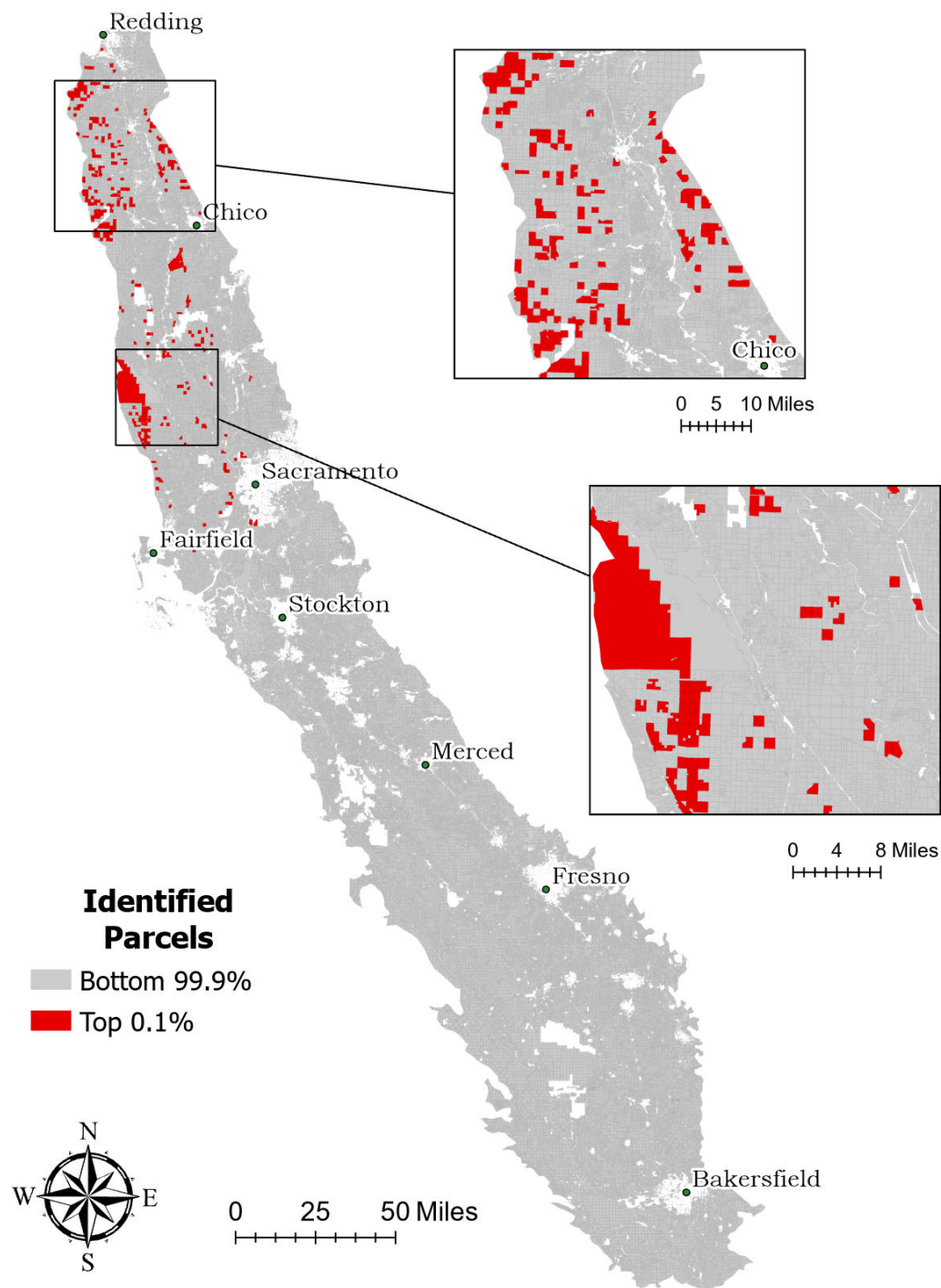
Notes: Map shows a simple solar suitability index (0-6) for non-urban parcels with no solar installation as of 2023, counting the number of favorable criteria each parcel meets. The six binary criteria are: (1) within 2km of an electrical substation, (2) larger than 40 hectares, (3) currently under low-value land use (fallow/idle cropland, grassland/pasture, developed land, or barren/shrubland), (4) in the top quartile of drought exposure, (5) above the 25th percentile of solar irradiance, and (6) in the bottom quartile of soil quality. Parcels scoring 5 or 6 represent high-priority sites combining strong solar potential with limited agricultural value. Parcels under low/medium/high developed land use are set to zero.

Figure A18: Comparison of Simple and Continuous Solar Suitability Measures



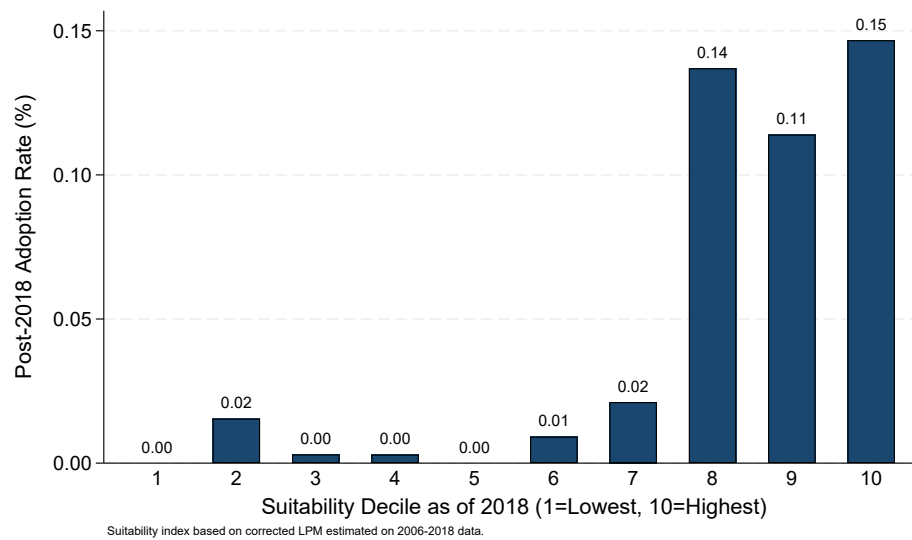
Note: Scatter plot comparing the simple count suitability index (0-6) with the continuous weighted suitability index (0-100) for parcels with no solar installations as of 2023. The simple index counts binary criteria (within 2km of substation, larger than 40ha, low-value land use, top quartile drought, above-median solar irradiance, bottom quartile soil quality). The continuous index applies linear penalty functions to the same underlying characteristics and weights them by standardized regression coefficients.

Figure A19: High-Priority Sites for Solar Development



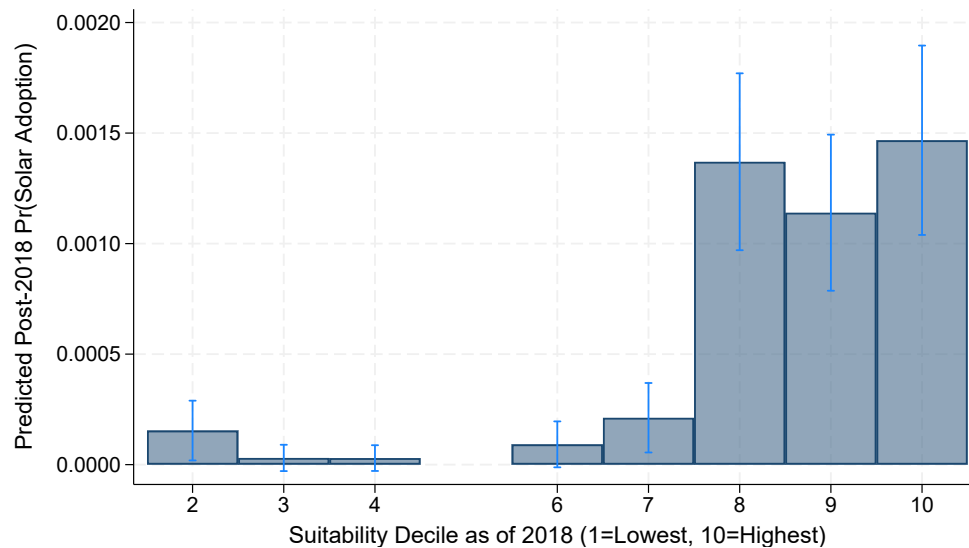
Note: Map shows 328 parcels scoring at or above the 99.9th percentile of the continuous solar suitability index among parcels that did not have solar installations as of 2023. These high-priority sites combine strong technical suitability for solar development (proximity to electrical infrastructure, large parcel size, high solar irradiance) with low agricultural value (high drought exposure, poor soil quality, low-value land uses).

Figure A20: Solar Adoption Rate in 2019-2023 by Predicted Suitability Decile



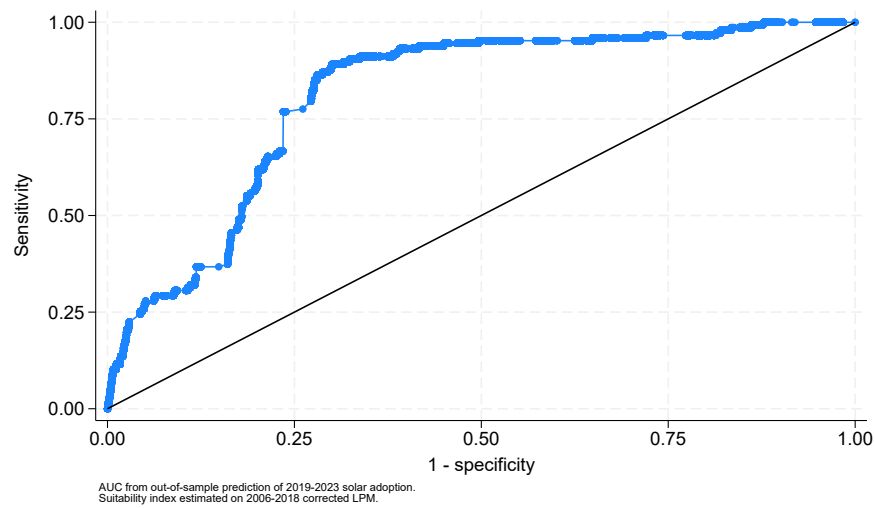
Note: Figure reports solar adoption rates for 2019-2023 by deciles of predicted suitability as of 2018, estimated using 2006-2018 data. Adoption rates indicate the percentage of parcels within each decile that adopted solar after 2018.

Figure A21: Predicted Probability of Post-2018 Solar Adoption by 2018 Suitability Decile



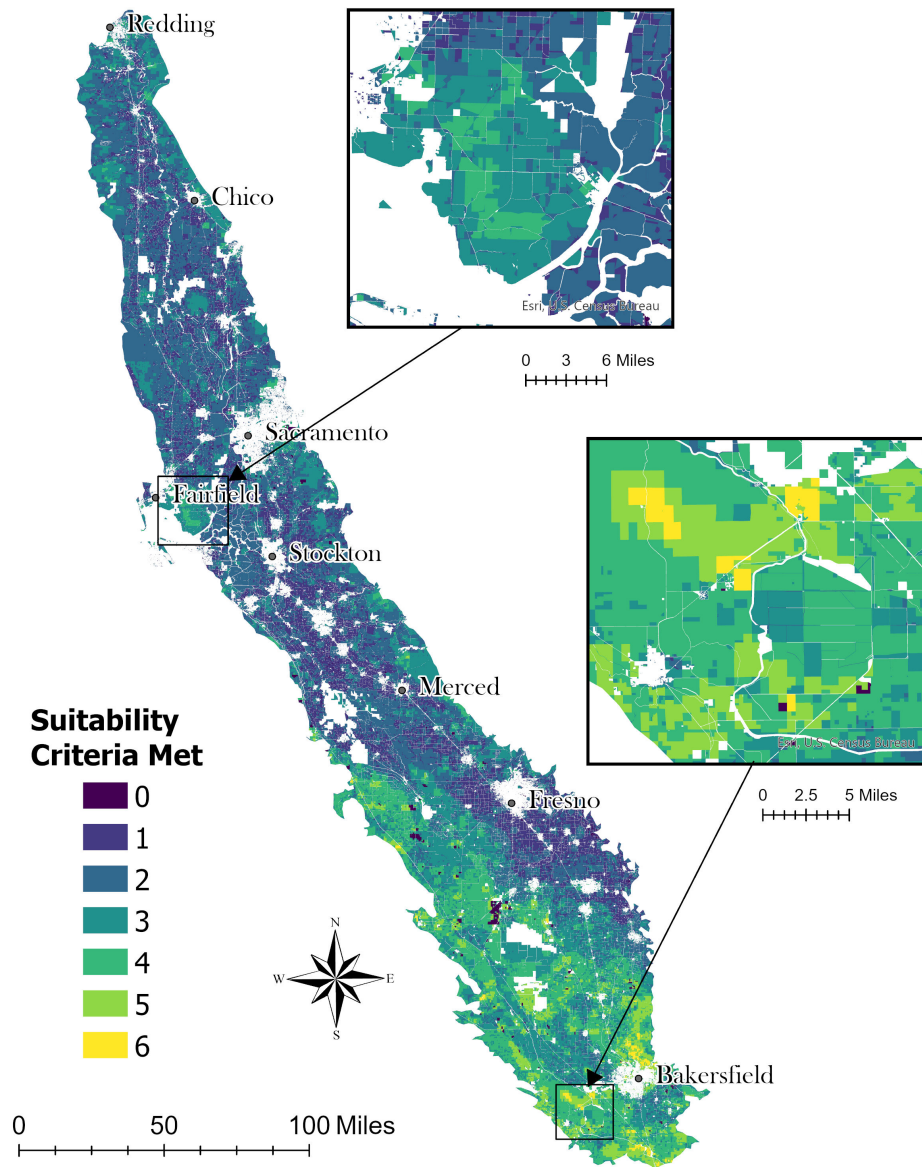
Note: Figure plots marginal effects from a logistic regression of post-2018 solar adoption on 2018 suitability deciles. The model uses suitability scores constructed from 2006-2018 data and applied to 2018 parcel characteristics.

Figure A22: ROC Curve for Suitability Index Prediction



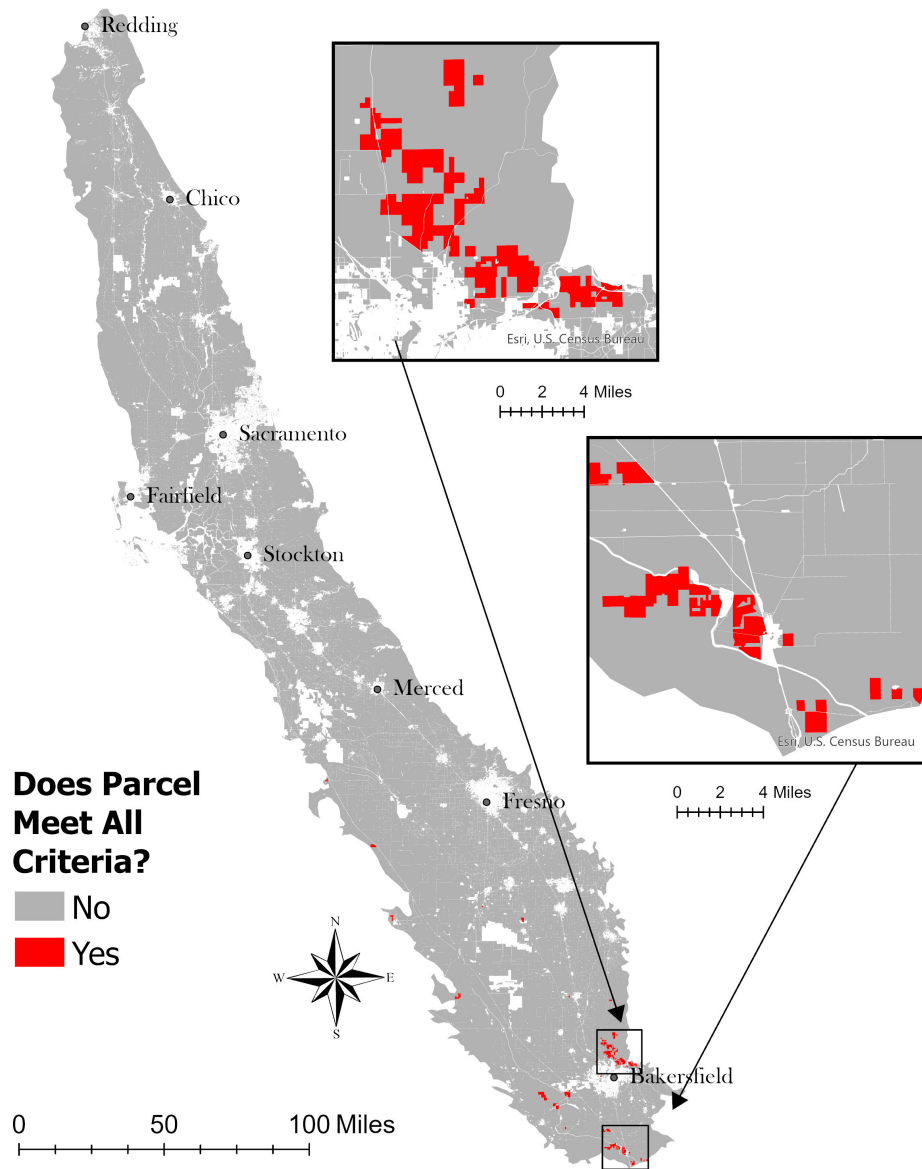
Note: Receiver operating characteristic (ROC) curve showing the predictive performance of the suitability index for 2019-2023 solar adoption. The index is constructed using coefficients estimated on 2006-2018 data and applied to 2018 parcel characteristics.

Figure A23: Solar Siting Suitability Score, Policy Criteria



Notes: Figure maps each never-solar parcel in the Central Valley, colored by the number of the following criteria it satisfied: well depth above the 90th percentile; not classified as Prime, Unique, or Important farmland; land cover other than water, developed, or forest/wetlands; solar irradiance above the 75th percentile; parcel area exceeding 40 hectares; and distance to the nearest electrical substation below 2km. Each criterion is coded as a binary indicator, and the score sums the number of criteria satisfied. Solar irradiance criterion drives prioritization toward the southern end of the Central Valley.

Figure A24: Highest-Priority Parcels, Policy Criteria



Notes: Figure highlights, in red, the parcels satisfying all six of the following criteria: well depth above the 90th percentile; not classified as Prime, Unique, or Important farmland; land cover other than water, developed, or forest/wetlands; solar irradiance above the 75th percentile; parcel area exceeding 40 hectares; and distance to the nearest electrical substation below 2km. Remaining parcels are shown in gray.